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(54) **LEFT ATRIAL APPENDAGE CLOSURE
DEVICE**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

178,283 A 6/1876 French
1,967,318 A 7/1934 Monahan
(Continued)

FOREIGN PATENT DOCUMENTS

CN 1399571 A 2/2003
CN 202143640 U 2/2012
(Continued)

OTHER PUBLICATIONS

International Search Report and Written Opinion dated Oct. 25,
2022 for International Application No. PCT/US2022/036092.

(Continued)

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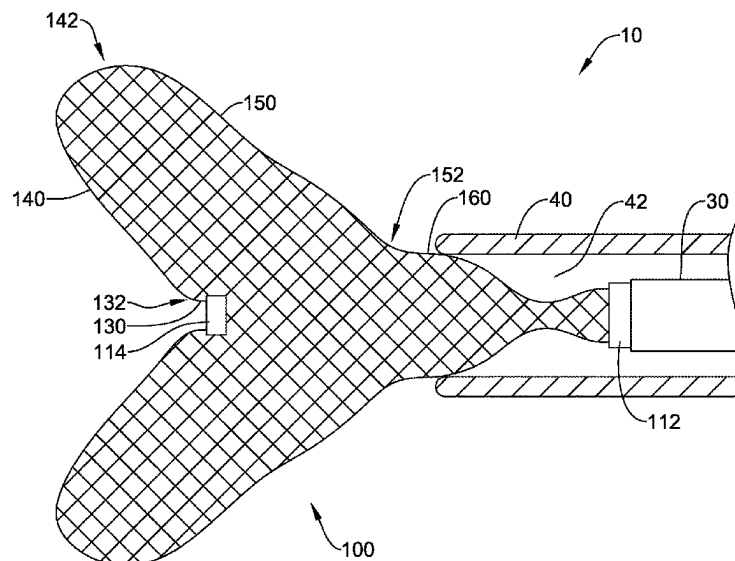
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(57) **ABSTRACT**

A left atrial appendage closure device may include an expandable framework having a plurality of struts joined at a proximal and distal hub. When the framework is fully constrained in a first position, a first segment of struts extends distally from the distal hub parallel to a central longitudinal axis to a first bend and a second segment of struts extends from the first bend proximally. A first amount of the framework is unconstrained in a second position, where the first segment extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment extends from the first bend proximally and radially outward to a second bend, a third segment of struts extends from the second bend proximally and radially inward to a third bend, and a fourth segment of struts extends from the third bend proximally to within the delivery sheath.

16 Claims, 9 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

3,402,710 A 9/1968 Paleschuck
 3,540,431 A 11/1970 Mobin-Uddin
 3,557,794 A 1/1971 Van Patten
 3,638,652 A 2/1972 Kelley
 3,811,449 A 5/1974 Gravlee et al.
 3,844,302 A 10/1974 Klein
 3,874,388 A 4/1975 King et al.
 4,007,743 A 2/1977 Blake
 4,108,420 A 8/1978 West et al.
 4,175,545 A 11/1979 Termanini
 4,309,776 A 1/1982 Berguer
 4,341,218 A 7/1982 Ü
 4,364,392 A 12/1982 Strother et al.
 4,425,908 A 1/1984 Simon
 4,545,367 A 10/1985 Tucci
 4,585,000 A 4/1986 Hershenson
 4,603,693 A 8/1986 Conta et al.
 4,611,594 A 9/1986 Grayhack et al.
 4,619,246 A 10/1986 Molgaard-Nielsen et al.
 4,638,803 A 1/1987 Rand et al.
 4,665,906 A 5/1987 Jervis
 4,681,588 A 7/1987 Ketharanathan et al.
 4,710,192 A 12/1987 Liotta et al.
 4,718,417 A 1/1988 Kittrell et al.
 4,759,348 A 7/1988 Cawood et al.
 4,781,177 A 11/1988 Lebigot
 4,793,348 A 12/1988 Palmaz
 4,827,907 A 5/1989 Tashiro
 4,832,055 A 5/1989 Palestiant
 4,873,978 A 10/1989 Ginsburg
 4,917,089 A 4/1990 Sideris
 4,921,484 A 5/1990 Hillstead
 4,960,412 A 10/1990 Fink
 4,966,150 A 10/1990 Etienne et al.
 4,998,972 A 3/1991 Chin et al.
 5,037,810 A 8/1991 Saliba, Jr.
 5,041,090 A 8/1991 Scheglov et al.
 5,041,093 A 8/1991 Chu
 5,042,707 A 8/1991 Taheri
 5,053,009 A 10/1991 Herzberg
 5,064,435 A 11/1991 Porter
 5,071,407 A 12/1991 Termin et al.
 5,078,736 A 1/1992 Behl
 5,098,440 A 3/1992 Hillstead
 5,108,418 A 4/1992 Lefebvre
 5,108,420 A 4/1992 Marks
 5,108,474 A 4/1992 Riedy et al.
 5,116,360 A 5/1992 Pinchuk et al.
 5,122,136 A 6/1992 Guglielmi et al.
 5,171,259 A 12/1992 Inoue
 5,171,383 A 12/1992 Sagaye et al.
 5,176,692 A 1/1993 Wilk et al.
 5,192,301 A 3/1993 Kamiya et al.
 5,211,658 A 5/1993 Clouse
 5,234,458 A 8/1993 Metais
 5,256,146 A 10/1993 Ensminger et al.
 5,258,000 A 11/1993 Gianturco
 5,258,042 A 11/1993 Mehta
 5,279,539 A 1/1994 Bohan et al.
 5,284,488 A 2/1994 Sideris
 5,304,184 A 4/1994 Hathaway et al.
 5,306,234 A 4/1994 Johnson
 5,312,341 A 5/1994 Turi
 5,329,942 A 7/1994 Gunther et al.
 5,334,217 A 8/1994 Das
 5,344,439 A 9/1994 Otten
 5,350,398 A 9/1994 Pavcnik et al.

5,350,399 A 9/1994 Erlebacher et al.
 5,353,784 A 10/1994 Nady-Mohamed
 5,366,460 A 11/1994 Eberbach
 5,366,504 A 11/1994 Andersen et al.
 5,370,657 A 12/1994 Irie
 5,375,612 A 12/1994 Cottenceau et al.
 5,397,331 A 3/1995 Himpens et al.
 5,397,355 A 3/1995 Marin et al.
 5,409,444 A 4/1995 Kensey et al.
 5,417,699 A 5/1995 Klein et al.
 5,421,832 A 6/1995 Lefebvre
 5,425,744 A 6/1995 Fagan et al.
 5,427,119 A 6/1995 Swartz et al.
 5,433,727 A 7/1995 Sideris
 5,443,454 A 8/1995 Tanabe et al.
 5,443,478 A 8/1995 Purdy et al.
 5,451,235 A 9/1995 Lock et al.
 5,454,365 A 10/1995 Bonutti
 5,464,408 A 11/1995 Duc
 5,469,867 A 11/1995 Schmitt
 5,490,856 A 2/1996 Person et al.
 5,497,774 A 3/1996 Swartz et al.
 5,499,975 A 3/1996 Cope et al.
 5,499,995 A 3/1996 Teirstein
 5,522,790 A 6/1996 Moll et al.
 5,522,822 A 6/1996 Phelps et al.
 5,522,836 A 6/1996 Palermo
 5,527,322 A 6/1996 Klein et al.
 5,527,338 A 6/1996 Purdy
 5,558,093 A 9/1996 Pomeranz et al.
 5,558,652 A 9/1996 Henke
 5,569,204 A 10/1996 Cramer et al.
 5,591,196 A 1/1997 Marin et al.
 5,614,204 A 3/1997 Cochrum
 5,634,936 A 6/1997 Linden et al.
 5,634,942 A 6/1997 Chevillon et al.
 5,637,097 A 6/1997 Yoon
 5,643,282 A 7/1997 Kieturakis
 5,643,292 A 7/1997 Hart
 5,649,953 A 7/1997 Lefebvre
 5,653,690 A 8/1997 Booth et al.
 5,662,671 A 9/1997 Barbut et al.
 5,669,933 A 9/1997 Simon et al.
 5,681,345 A 10/1997 Euteneuer
 5,681,347 A 10/1997 Cathcart et al.
 5,683,411 A 11/1997 Kavteladze et al.
 5,690,671 A 11/1997 McGurk et al.
 5,693,067 A 12/1997 Purdy
 5,695,525 A 12/1997 Mulhauser et al.
 5,700,285 A 12/1997 Myers et al.
 5,702,421 A 12/1997 Schneidt
 5,704,910 A 1/1998 Humes
 5,709,224 A 1/1998 Behl et al.
 5,709,704 A 1/1998 Nott et al.
 5,709,707 A 1/1998 Lock et al.
 5,722,400 A 3/1998 Ockuly et al.
 5,724,975 A 3/1998 Negus et al.
 5,725,512 A 3/1998 Swartz et al.
 5,725,552 A 3/1998 Kotula et al.
 5,725,568 A 3/1998 Hastings
 5,733,294 A 3/1998 Forber et al.
 5,733,302 A 3/1998 Myler et al.
 5,735,290 A 4/1998 Sterman et al.
 5,749,880 A 5/1998 Banas et al.
 5,749,883 A 5/1998 Halpern
 5,749,894 A 5/1998 Engelson
 5,766,219 A 6/1998 Horton
 5,766,246 A 6/1998 Mulhauser et al.
 5,769,816 A 6/1998 Barbut et al.
 5,776,097 A 7/1998 Massoud
 5,776,162 A 7/1998 Kleshinski
 5,782,860 A 7/1998 Epstein et al.
 5,785,679 A 7/1998 Abolfathi et al.
 5,800,454 A 9/1998 Jacobsen et al.
 5,800,457 A 9/1998 Gelbfish
 5,800,512 A 9/1998 Letnz et al.
 5,807,261 A 9/1998 Benaron et al.
 5,810,874 A 9/1998 Lefebvre
 5,814,028 A 9/1998 Swartz et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,814,029	A	9/1998	Hassett	6,036,720	A	3/2000	Abrams et al.
5,814,064	A	9/1998	Daniel	6,042,598	A	3/2000	Tsugita et al.
5,820,591	A	10/1998	Thompson et al.	6,048,331	A	4/2000	Tsugita et al.
5,823,198	A	10/1998	Jones et al.	6,051,014	A	4/2000	Jang
5,830,228	A	11/1998	Knapp et al.	6,051,015	A	4/2000	Maahs
5,833,673	A	11/1998	Ockuly et al.	6,056,720	A	5/2000	Morse
5,836,913	A	11/1998	Orth et al.	6,063,070	A	5/2000	Eder
5,836,968	A	11/1998	Simon et al.	6,063,113	A	5/2000	Kavteladze et al.
5,840,027	A	11/1998	Swartz et al.	6,066,126	A	5/2000	Li et al.
5,843,118	A	12/1998	Sepetka et al.	6,068,621	A	5/2000	Balceta et al.
5,846,260	A	12/1998	Maahs	6,074,357	A	6/2000	Kaganov et al.
5,846,261	A	12/1998	Kotula et al.	6,076,012	A	6/2000	Swanson et al.
5,848,969	A	12/1998	Panescu et al.	6,079,414	A	6/2000	Roth
5,849,005	A	12/1998	Garrison et al.	6,080,182	A	6/2000	Shaw et al.
5,851,232	A	12/1998	Lois	6,080,183	A	6/2000	Tsugita et al.
5,853,422	A	12/1998	Huebsch et al.	6,083,239	A	7/2000	Addis
5,855,597	A	1/1999	Jayaraman	6,090,084	A	7/2000	Hassett et al.
5,865,791	A	2/1999	Whayne et al.	6,096,052	A	8/2000	Callister et al.
5,865,802	A	2/1999	Yoon et al.	6,096,053	A	8/2000	Bates et al.
5,868,702	A	2/1999	Stevens et al.	6,110,243	A	8/2000	Wnenschak et al.
5,868,708	A	2/1999	Hart et al.	6,123,715	A	9/2000	Amplatz
5,876,367	A	3/1999	Kaganov et al.	6,124,523	A	9/2000	Banas et al.
5,879,296	A	3/1999	Ockuly et al.	6,132,438	A	10/2000	Fleischman et al.
5,879,366	A	3/1999	Shaw et al.	6,135,991	A	10/2000	Muni et al.
5,882,340	A	3/1999	Yoon	6,136,016	A	10/2000	Barbut et al.
5,885,258	A	3/1999	Sachdeva et al.	6,139,527	A	10/2000	Laufer et al.
5,891,558	A	4/1999	Bell et al.	6,139,573	A	10/2000	Sogard et al.
5,895,399	A	4/1999	Barbut et al.	6,152,144	A	11/2000	Lesh et al.
5,902,289	A	5/1999	Swartz et al.	6,152,946	A	11/2000	Broome et al.
5,904,680	A	5/1999	Kordis et al.	6,156,055	A	12/2000	Ravenscroft
5,904,703	A	5/1999	Gilson	6,159,195	A	12/2000	Ha et al.
5,906,207	A	5/1999	Shen	6,161,543	A	12/2000	Cox et al.
5,910,154	A	6/1999	Tsugita et al.	6,168,615	B1	1/2001	Ken et al.
5,911,734	A	6/1999	Tsugita et al.	6,171,329	B1	1/2001	Shaw et al.
5,916,236	A	6/1999	Muij Van de Moer et al.	6,179,859	B1	1/2001	Bates et al.
5,925,060	A	7/1999	Forber	6,193,739	B1	2/2001	Chevillon et al.
5,925,063	A	7/1999	Khosravi	6,203,531	B1	3/2001	Ockuly et al.
5,925,074	A	7/1999	Gingras et al.	6,206,907	B1	3/2001	Marino et al.
5,925,075	A	7/1999	Myers et al.	6,214,029	B1	4/2001	Thill et al.
5,928,192	A	7/1999	Maahs	6,221,092	B1	4/2001	Koike et al.
5,928,260	A	7/1999	Chin et al.	6,231,561	B1	5/2001	Frazier et al.
5,931,818	A	8/1999	Werp et al.	6,231,589	B1	5/2001	Wessman et al.
5,935,145	A	8/1999	Villar et al.	6,235,045	B1	5/2001	Barbut et al.
5,935,147	A	8/1999	Kensey et al.	6,245,012	B1	6/2001	Kleshinski
5,935,148	A	8/1999	Villar et al.	6,251,122	B1	6/2001	Tsukernik
5,941,249	A	8/1999	Maynard	6,258,115	B1	7/2001	Dubrul
5,941,896	A	8/1999	Kerr	6,267,772	B1	7/2001	Mulhauser et al.
5,944,738	A	8/1999	Amplatz et al.	6,267,776	B1	7/2001	O'Connell
5,947,997	A	9/1999	Pavcnik et al.	6,270,490	B1	8/2001	Hahnen
5,951,589	A	9/1999	Epstein et al.	6,270,530	B1	8/2001	Eldridge et al.
5,951,599	A	9/1999	McCrorry	6,270,902	B1	8/2001	Tedeschi et al.
5,954,694	A	9/1999	Sunseri	6,277,138	B1	8/2001	Levinson et al.
5,954,767	A	9/1999	Pajotin et al.	6,285,898	B1	9/2001	Ben-Haim
5,957,940	A	9/1999	Tanner et al.	6,290,674	B1	9/2001	Roue et al.
5,961,545	A	10/1999	Lentz et al.	6,290,708	B1	9/2001	Kugel et al.
5,976,174	A	11/1999	Ruiz	6,312,407	B1	11/2001	Zadno-Azizi et al.
5,980,514	A	11/1999	Kupiecki et al.	6,319,251	B1	11/2001	Tu et al.
5,980,555	A	11/1999	Barbut et al.	6,328,727	B1	12/2001	Frazier et al.
5,989,281	A	11/1999	Barbut et al.	6,328,755	B1	12/2001	Marshall
5,993,469	A	11/1999	McKenzie et al.	6,342,062	B1	1/2002	Suon et al.
5,993,483	A	11/1999	Gianotti	6,346,116	B1	2/2002	Brooks et al.
5,997,557	A	12/1999	Barbut et al.	6,346,895	B1	2/2002	Lee et al.
6,004,280	A	12/1999	Buck et al.	6,361,545	B1	3/2002	Macoviak et al.
6,004,348	A	12/1999	Banas et al.	6,364,895	B1	4/2002	Greenhalgh
6,007,523	A	12/1999	Mangosong	6,368,338	B1	4/2002	Konya et al.
6,007,557	A	12/1999	Ambrisco et al.	6,371,971	B1	4/2002	Tsugita et al.
6,010,517	A	1/2000	Baccaro	6,375,670	B1	4/2002	Greenhalgh
6,010,522	A	1/2000	Barbut et al.	6,391,044	B1	5/2002	Yadav et al.
6,013,093	A	1/2000	Nott et al.	6,398,803	B1	6/2002	Layne et al.
6,024,751	A	2/2000	Lovato et al.	6,402,746	B1	6/2002	Whayne et al.
6,024,754	A	2/2000	Engelson	6,402,771	B1	6/2002	Palmer et al.
6,024,755	A	2/2000	Addis	6,402,779	B1	6/2002	Colone et al.
6,024,756	A	2/2000	Huebsch et al.	6,419,669	B1	7/2002	Frazier et al.
6,027,520	A	2/2000	Tsugita et al.	6,440,152	B1	8/2002	Gainor et al.
6,033,420	A	3/2000	Hahnen	6,443,972	B1	9/2002	Bosma et al.
				6,447,530	B1	9/2002	Ostrovsky et al.
				6,454,775	B1	9/2002	Demarais et al.
				6,458,145	B1	10/2002	Ravenscroft et al.
				6,464,712	B1	10/2002	Epstein et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,468,291 B2	10/2002	Bates et al.	8,080,032 B2	12/2011	van der Burg et al.
6,468,301 B1	10/2002	Amplatz et al.	8,097,015 B2	1/2012	Devellian
6,485,501 B1	11/2002	Green	8,100,938 B2	1/2012	Figulla et al.
6,488,689 B1	12/2002	Kaplan et al.	8,221,384 B2	7/2012	Frazier et al.
6,511,496 B1	1/2003	Huter et al.	8,221,445 B2	7/2012	van Tassel et al.
6,514,280 B1	2/2003	Gilson	8,287,563 B2	10/2012	Khairkhahan et al.
6,517,573 B1	2/2003	Pollock et al.	8,323,309 B2	12/2012	Khairkhahan et al.
6,533,782 B2	3/2003	Howell et al.	8,388,672 B2	3/2013	Khairkhahan et al.
6,547,760 B1	4/2003	Samson et al.	8,491,623 B2	7/2013	Vogel et al.
6,547,815 B2	4/2003	Myers	8,523,897 B2	9/2013	van der Burg et al.
6,551,303 B1	4/2003	Van Tassel et al.	8,535,343 B2	9/2013	van der Burg et al.
6,551,344 B2	4/2003	Thill	8,562,509 B2	10/2013	Bates
6,558,401 B1	5/2003	Azizi	8,663,273 B2	3/2014	Khairkhahan et al.
6,558,405 B1	5/2003	McInnes	8,685,055 B2	4/2014	VanTassel et al.
6,558,414 B2	5/2003	Layne	8,728,117 B1	5/2014	Janardhan et al.
6,562,058 B2	5/2003	Seguin et al.	8,758,389 B2	6/2014	Glimsdale
6,569,184 B2	5/2003	Huter	8,828,051 B2	9/2014	Javois et al.
6,569,214 B2	5/2003	Williams et al.	8,834,519 B2	9/2014	van der Burg et al.
6,589,214 B2	7/2003	McGuckin et al.	8,845,711 B2	9/2014	Miles et al.
6,589,251 B2	7/2003	Yee et al.	9,034,006 B2	5/2015	Quinn et al.
6,599,308 B2	7/2003	Amplatz	9,132,000 B2	9/2015	VanTassel et al.
6,602,271 B2	8/2003	Adams et al.	9,168,043 B2	10/2015	van der Burg et al.
6,623,508 B2	9/2003	Shaw et al.	9,211,124 B2	12/2015	Campbell et al.
6,641,564 B1	11/2003	Kraus	9,295,472 B2	3/2016	Ottma
6,650,923 B1	11/2003	Lesh et al.	9,351,716 B2	5/2016	Miles et al.
6,652,555 B1	11/2003	VanTassel et al.	9,445,895 B2	9/2016	Kreidler
6,652,556 B1	11/2003	VanTassel et al.	9,554,804 B2	1/2017	Erzberger et al.
6,666,861 B1	12/2003	Grabek	9,554,806 B2	1/2017	Larsen et al.
6,689,150 B1	2/2004	Vantassel et al.	9,561,037 B2	2/2017	Fogarty et al.
6,699,260 B2	3/2004	Dubrul et al.	9,561,097 B1	2/2017	Kim et al.
6,699,276 B2	3/2004	Sogard et al.	9,592,058 B2	3/2017	Erzberger et al.
6,702,825 B2	3/2004	Frazier et al.	9,597,088 B2	3/2017	Ottma
6,712,836 B1	3/2004	Berg et al.	9,629,636 B2	4/2017	Fogarty et al.
6,726,701 B2	4/2004	Gilson et al.	9,730,701 B2	8/2017	Tischler et al.
6,730,108 B2	5/2004	Van Tassel et al.	9,750,505 B2	9/2017	Miles et al.
6,755,812 B2	6/2004	Peterson et al.	9,763,666 B2	9/2017	Wu et al.
6,827,737 B2	12/2004	Hill et al.	9,795,387 B2	10/2017	Miles et al.
6,837,901 B2	1/2005	Rabkin et al.	9,808,253 B2	11/2017	Li et al.
6,855,153 B2	2/2005	Saadat	9,883,936 B2	2/2018	Sutton et al.
6,911,037 B2	6/2005	Gainor et al.	9,913,652 B2	3/2018	Bridgeman et al.
6,932,838 B2	8/2005	Schwartz et al.	9,943,299 B2	4/2018	Khairkhahan et al.
6,942,653 B2	9/2005	Quinn	9,943,315 B2	4/2018	Kaplan et al.
6,949,113 B2	9/2005	Van Tassel et al.	10,071,181 B1	9/2018	Penegor et al.
6,958,061 B2	10/2005	Truckai et al.	10,076,335 B2	9/2018	Zaver et al.
6,994,092 B2	2/2006	van der Burg et al.	10,143,458 B2	12/2018	Kreidler
7,011,671 B2	3/2006	Welch	10,201,337 B2	2/2019	Glimsdale
7,014,645 B2	3/2006	Greene, Jr. et al.	10,231,737 B2	3/2019	Amplatz et al.
7,037,321 B2	5/2006	Sachdeva et al.	2001/0000797 A1	5/2001	Mazzocchi
7,044,134 B2	5/2006	Khairkhahan et al.	2001/0020181 A1	9/2001	Layne
7,097,651 B2	8/2006	Harrison et al.	2001/0034537 A1	10/2001	Shaw et al.
7,128,073 B1	10/2006	van der Burg et al.	2001/0037141 A1	11/2001	Yee et al.
7,152,605 B2	12/2006	Khairkhahan et al.	2002/0022860 A1	2/2002	Borillo et al.
7,169,164 B2	1/2007	Borillo et al.	2002/0035374 A1	3/2002	Borillo et al.
7,179,275 B2	2/2007	McGuckin, Jr. et al.	2002/0045931 A1	4/2002	Sogard et al.
7,226,466 B2	6/2007	Opolski	2002/0062133 A1	5/2002	Gilson et al.
7,303,526 B2	12/2007	Sharkey et al.	2002/0082638 A1	6/2002	Porter et al.
7,323,002 B2	1/2008	Johnson et al.	2002/0082675 A1	6/2002	Myers
7,597,704 B2	10/2009	Frazier et al.	2002/0099439 A1	7/2002	Schwartz et al.
7,678,123 B2	3/2010	Chanduszko	2002/0111647 A1	8/2002	Khairkhahan et al.
7,695,425 B2	4/2010	Schweich et al.	2002/0138094 A1	9/2002	Borillo et al.
7,713,282 B2	5/2010	Frazier et al.	2002/0138097 A1	9/2002	Ostrovsky et al.
7,722,641 B2	5/2010	van der Burg et al.	2002/0169475 A1	11/2002	Gainor et al.
7,727,189 B2	6/2010	VanTassel et al.	2002/0177855 A1	11/2002	Greene, Jr. et al.
7,735,493 B2	6/2010	van der Burg et al.	2003/0017775 A1	1/2003	Dong et al.
7,780,694 B2	8/2010	Palmer et al.	2003/0023262 A1	1/2003	Welch
7,799,049 B2	9/2010	Ostrovsky et al.	2003/0023266 A1	1/2003	Borillo et al.
7,811,300 B2	10/2010	Feller, III et al.	2003/0057156 A1	3/2003	Peterson et al.
7,811,314 B2	10/2010	Fierens et al.	2003/0060871 A1	3/2003	Hill et al.
7,862,500 B2	1/2011	Khairkhahan et al.	2003/0120337 A1	6/2003	Van Tassel et al.
7,927,365 B2	4/2011	Fierens et al.	2003/0181942 A1	9/2003	Sutton et al.
7,972,359 B2	7/2011	Kreidler	2003/0191526 A1	10/2003	Van Tassel et al.
8,025,495 B2	9/2011	Hardert et al.	2003/0195555 A1	10/2003	Khairkhahan et al.
8,043,329 B2	10/2011	Khairkhahan et al.	2003/0204203 A1	10/2003	Khairkhahan et al.
8,052,715 B2	11/2011	Quinn et al.	2003/0208214 A1	11/2003	Loshakove et al.
8,062,282 B2	11/2011	Kolb	2003/0220667 A1	11/2003	van der Burg et al.
			2004/0034366 A1	2/2004	van der Burg et al.
			2004/0049210 A1	3/2004	VanTassel et al.
			2004/0093012 A1	5/2004	Cully et al.
			2004/0098031 A1	5/2004	van der Burg et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0122467	A1	6/2004	VanTassel et al.
2004/0127935	A1	7/2004	VanTassel et al.
2004/0158274	A1	8/2004	WasDyke
2004/0186486	A1	9/2004	Roue et al.
2004/0215230	A1	10/2004	Frazier et al.
2004/0220610	A1	11/2004	Kreidler et al.
2004/0220682	A1	11/2004	Levine et al.
2004/0230222	A1	11/2004	van der Burg et al.
2005/0004652	A1	1/2005	van der Burg et al.
2005/0015109	A1	1/2005	Lichtenstein
2005/0038470	A1	2/2005	van der Burg et al.
2005/0049573	A1	3/2005	Van Tassel et al.
2005/0070952	A1	3/2005	Devellian
2005/0113861	A1	5/2005	Corcoran et al.
2005/0125020	A1	6/2005	Meade et al.
2005/0177182	A1	8/2005	van der Burg et al.
2005/0203568	A1	9/2005	Burg et al.
2005/0283186	A1	12/2005	Berrada et al.
2005/0288704	A1	12/2005	Cartier et al.
2006/0015136	A1	1/2006	Besselink
2006/0030877	A1	2/2006	Martinez et al.
2006/0052816	A1	3/2006	Bates et al.
2006/0100658	A1	5/2006	Obana et al.
2006/0155323	A1	7/2006	Porter et al.
2007/0066993	A1	3/2007	Kreidler
2007/0083227	A1	4/2007	van der Burg et al.
2007/0083230	A1	4/2007	Javois
2007/0150041	A1	6/2007	Evans et al.
2007/0156123	A1	7/2007	Moll et al.
2007/0162048	A1	7/2007	Quinn et al.
2007/0185471	A1	8/2007	Johnson
2008/0275536	A1	11/2008	Zarins et al.
2009/0005803	A1	1/2009	Batiste
2009/0062841	A1	3/2009	Amplatz et al.
2009/0099647	A1	4/2009	Glimsdale et al.
2009/0105747	A1	4/2009	Chanduszko et al.
2009/0112249	A1	4/2009	Miles et al.
2009/0254195	A1	10/2009	Khairkhan et al.
2009/0318948	A1	12/2009	Linder et al.
2010/0004726	A1	1/2010	Hancock et al.
2010/0049238	A1	2/2010	Simpson
2010/0106178	A1	4/2010	Obermiller et al.
2010/0324585	A1	12/2010	Miles et al.
2011/0054515	A1*	3/2011	Bridgeman A61B 17/12122
			606/200
2011/0082495	A1	4/2011	Ruiz
2011/0098525	A1	4/2011	Kermode et al.
2011/0218566	A1	9/2011	van der Burg et al.
2011/0301630	A1	12/2011	Hendriksen et al.
2012/0029553	A1	2/2012	Quinn et al.
2012/0035643	A1	2/2012	Khairkahan et al.
2012/0065662	A1	3/2012	van der Burg et al.
2012/0125619	A1	5/2012	Wood et al.
2012/0172654	A1	7/2012	Bates
2012/0172927	A1	7/2012	Campbell et al.
2012/0239077	A1	9/2012	Zaver et al.
2012/0239083	A1	9/2012	Kreidler
2012/0245619	A1	9/2012	Guest
2012/0283585	A1	11/2012	Werneth et al.
2012/0283773	A1	11/2012	Van Tassel et al.
2012/0323267	A1	12/2012	Ren
2013/0006343	A1	1/2013	Kassab et al.
2013/0012982	A1	1/2013	Khairkahan et al.
2013/0018413	A1	1/2013	Oral et al.
2013/0110154	A1	5/2013	van der Burg et al.
2013/0165735	A1	6/2013	Khairkahan et al.
2013/0211492	A1	8/2013	Schneider et al.
2013/0331884	A1	12/2013	Van der Burg et al.
2014/0005714	A1	1/2014	Quick et al.
2014/0018841	A1	1/2014	Peiffer et al.
2014/0039536	A1	2/2014	Cully et al.
2014/0046360	A1	2/2014	van der Burg et al.
2014/0081314	A1	3/2014	Zaver et al.
2014/0100596	A1	4/2014	Rudman et al.
2014/0135817	A1	5/2014	Tischler et al.
2014/0142612	A1	5/2014	Li et al.
2014/0148842	A1	5/2014	Khairkahan et al.
2014/0163605	A1	6/2014	VanTassel et al.
2014/0188157	A1	7/2014	Clark
2014/0214077	A1	7/2014	Glimsdale
2014/0296908	A1	10/2014	Ottma et al.
2014/0303719	A1	10/2014	Cox et al.
2014/0336612	A1	11/2014	Frydlewski et al.
2014/0336699	A1	11/2014	van der Burg et al.
2015/0005810	A1	1/2015	Center et al.
2015/0039021	A1	2/2015	Khairkahan et al.
2015/0080903	A1	3/2015	Dillard et al.
2015/0196300	A1	7/2015	Tischler et al.
2015/0230909	A1	8/2015	Zaver et al.
2015/0238197	A1	8/2015	Quinn et al.
2015/0305727	A1	10/2015	Karimov et al.
2015/0313604	A1	11/2015	Roue et al.
2015/0313605	A1	11/2015	Griffin
2015/0327979	A1	11/2015	Quinn et al.
2015/0374491	A1	12/2015	Kreidler
2016/0015397	A1	1/2016	Figulla et al.
2016/0051358	A1	2/2016	Sutton et al.
2016/0058539	A1	3/2016	VanTassel et al.
2016/0066922	A1	3/2016	Bridgeman et al.
2016/0106437	A1	4/2016	van der Burg et al.
2016/0192942	A1	7/2016	Strauss et al.
2016/0287259	A1	10/2016	Hanson et al.
2016/0331382	A1	11/2016	Center et al.
2016/0374657	A1	12/2016	Kreidler
2017/0007262	A1	1/2017	Amplatz et al.
2017/0027552	A1	2/2017	Turkington et al.
2017/0042550	A1	2/2017	Chakraborty et al.
2017/0056166	A1	3/2017	Ratz et al.
2017/0100112	A1	4/2017	van der Burg et al.
2017/0119400	A1	5/2017	Amplatz et al.
2017/0181751	A1	6/2017	Larsen et al.
2017/0224354	A1	8/2017	Tischler et al.
2017/0340336	A1	11/2017	Osycka
2017/0354421	A1	12/2017	Maguire et al.
2018/0064446	A1	3/2018	Figulla et al.
2018/0070950	A1	3/2018	Zaver et al.
2018/0140412	A1	5/2018	Sutton et al.
2018/0140413	A1	5/2018	Quinn et al.
2018/0250014	A1	9/2018	Melanson et al.
2019/0133563	A1	5/2019	Glimsdale
2019/0175185	A1	6/2019	Amplatz et al.
2019/0223883	A1	7/2019	Anderson et al.
2019/0247053	A1	8/2019	Inouye
2021/0059685	A1	3/2021	Groff et al.

FOREIGN PATENT DOCUMENTS

CN	104287804	A	1/2015
CN	1104352261	A	2/2015
CN	106859722	A	6/2017
CN	1109464173	A	3/2019
DE	10201004476	A1	3/2012
EP	1523957	A2	4/2005
EP	1595504	A1	11/2005
EP	2074953	A1	1/2009
EP	2481381	A1	8/2012
EP	2928420	A1	10/2015
EP	3072461	A1	9/2016
EP	3372173	A2	9/2018
EP	3398523	A1	11/2018
JP	2003532457	A	11/2003
JP	2005324019	A	11/2005
JP	2007513684	A	5/2007
JP	2009160402	A	7/2009
JP	2012501793	A	1/2012
WO	9313712	A1	7/1993
WO	9504132	A1	2/1995
WO	9522359	A1	8/1995
WO	9601591	A1	1/1996
WO	9640356	A1	12/1996
WO	9721402	A1	6/1997
WO	9726939	A1	7/1997
WO	9728749	A1	8/1997
WO	9735522	A1	10/1997

(56)

References Cited**FOREIGN PATENT DOCUMENTS**

WO	9802100	A1	1/1998
WO	9817187	A1	4/1998
WO	9822026	A1	5/1998
WO	9823322	A1	6/1998
WO	9827868	A1	7/1998
WO	9905977	A1	2/1999
WO	9907289	A1	2/1999
WO	9908607	A1	2/1999
WO	9923976	A1	5/1999
WO	9925252	A1	5/1999
WO	9930640	A1	6/1999
WO	9944510	A1	9/1999
WO	9959479	A1	11/1999
WO	0001308	A1	1/2000
WO	0016705	A1	3/2000
WO	0027292	A1	5/2000
WO	0035352	A1	6/2000
WO	0053120	A1	9/2000
WO	0067669	A1	11/2000
WO	0108743	A1	2/2001
WO	0115629	A1	3/2001
WO	0121247	A1	3/2001
WO	0126726	A1	4/2001
WO	0130266	A1	5/2001
WO	0130267	A1	5/2001
WO	0130268	A1	5/2001
WO	0170119	A1	9/2001
WO	0215793	A2	2/2002
WO	0224106	A2	3/2002
WO	02071977	A2	9/2002
WO	03007825	A1	1/2003
WO	03008030	A2	1/2003
WO	03032818	A1	4/2003
WO	2004012629	A1	2/2004
WO	2007044536	A1	4/2007
WO	2010024801	A1	3/2010
WO	2010081033	A1	7/2010
WO	2013060855	A1	5/2013
WO	2013159065	A1	10/2013
WO	2014011865	A1	1/2014
WO	2014018907	A1	1/2014
WO	2014089129	A1	6/2014
WO	201406239	A1	7/2014
WO	2015164836	A1	10/2015
WO	2016087145	A1	6/2016
WO	2018017935	A1	1/2018
WO	2018187732	A1	10/2018
WO	2019084358	A1	5/2019

OTHER PUBLICATIONS

International Search Report and Written Opinion dated Aug. 3, 2004 for International Application No. PCT/US2004/008109.

International Search Report and Written Opinion dated Feb. 15, 2000 for International Application No. PCT/US99/26325.

International Search Report dated May 20, 2003 for International Application No. PCT/US02/33808.

Written Opinion dated Nov. 17, 2003 for International Application No. PCT/US/02/33808.

International Search Report and Written Opinion dated Aug. 21, 2018 for International Application No. PCT/US2018/029684.

Cragg et al., "A New Percutaneous Vena Cava Filter," American Journal of Radiology, Sep. 1983, pp. 601-604, vol. 141.

Cragg et al., "Nonsurgical Placement of Arterial Endoprostheses: A New Technique Using Nitinol Wire," Radiology, Apr. 1983, pp. 261-263, vol. 147, No. 1.

Lock et al., "Transcatheter Closure of Atrial Septal Defects," Circulation, May 1989, pp. 1091-1099, vol. 79, No. 5.

Lock et al., "Transcatheter Umbrella Closure of Congenital Heart Defects," Circulation, Mar. 1987, pp. 593-599, vol. 75, No. 3.

Rashkind et al., "Nonsurgical closure of patent ductus arteriosus: clinical application of the Rashkind PDA Occluder System," Circulation, Mar. 1987, pp. 583-592, vol. 75, No. 3.

Rosengart et al., "Percutaneous and Minimally Invasive Valve Procedures," Circulation, Apr. 1, 2008, pp. 1750-1767, vol. 117.

Ruttenberg, "Nonsurgical Therapy of Cardiac Disorders," Pediatric Consult, 1986, Pages not numbered, vol. 5, No. 2.

Sugita et al., "Nonsurgical Implantations of a Vascular Ring Prosthesis Using Thermal Shape Memory Ti/Ni Alloy (Nitinol Wire)," Trans. Am. Soc. Artif. Intern. Organs, 1986, pp. 30-34, vol. XXXII.

Wessel et al., "Outpatient Closure of the Patent Ductus Arteriosus," Circulation, 1988, pp. 1068-1071, vol. 77, No. 5.

Tung et al., U.S. Appl. No. 61/559,941, filed Nov. 15, 2011.

Yue Yu et al., U.S. Appl. No. 61/557,880, filed Dec. 20, 2011.

Cline, "File: Fish hooks.jpg," Wikipedia foundation, Inc., San Francisco, CA, Jun. 2007; p. 1 of 4; available online at http://en.wikipedia.org/wiki/File:Fish_hooks.jpg; last accessed Oct. 5, 2012.

International Search Report and Written Opinion dated Apr. 22, 2014 for International Application No. PCT/US2013/078454.

Aryana et al., "Incomplete Closure of the Left Atrial Appendage: Implication and Management," Curr Cardiol Rep., 18(9):82, 2016.

Delurgio, "Device-Associated Thrombus and Peri-Device Leak Following Left Atrial Appendage Closure with the Amplatzer Cardiac Plug," JACC: Cardiovascular Interventions, 10(4): 400-402, 2017.

University of Minnesota. Atlas of Human Cardiac Anatomy, Left Atrium. Retrieved from <http://www.vhlab.umn.edu/atlas/left-atrium/left-atrial-appendage/index.shtml>. Accessed 2017. Downloaded 2019.

Saw et al., "Incidence and Clinical Impact of Device-Associated Thrombus and Peri-Device Leak following Left Atrial Appendage Closure with the Amplatzer Cardiac Plug," JACC: Cardiovascular Intervention. 10(4): 391-399, 2017.

Romero et al., "Left Atrial Appendage Closure Devices," Clinical Medicine Insights: Cardiology, vol. 8, pp. 45-52, 2014.

Invitation To Pay Additional Fees And, Where Applicable, Protest Fee, mailed Oct. 13, 2016.

International Search Report and Written Opinion dated Oct. 14, 2019 for International Application No. PCT/US2019/047452.

International Search Report and Written Opinion dated Oct. 27, 2017 for International Application No. PCT/US2017/048150.

International Search Report and Written Opinion dated Jan. 21, 2019 for International Application No. PCT/US2018/051953.

International Search Report and Written Opinion dated Oct. 13, 2016 for International Application No. PCT/US2016/043363.

International Search Report and Written Opinion dated Mar. 17, 2020, for International Application No. PCT/US2019/065243.

International Search Report and Written Opinion dated Sep. 9, 2019 for International Application No. PCT/US2019/033698.

Blackshear et al.; "Appendage Obliteration to Reduce Stroke in Cardiac Surgical Patients with Atrial Fibrillation", Ann. Thoracic Surgery, pp. 755-759, 1996.

Lindsay, "Obliteration of the Left Atrial Appendage: A Concept Worth Testing", Ann. Thoracic Surgery, 1996.

Invitation To Pay Additional Fees dated Feb. 22, 2019 for International Application No. PCT/US2018/066163.

International Search Report and Written Opinion dated Oct. 20, 2020 for International Application No. PCT/US2020/042192.

International Search Report and Written Opinion dated Oct. 23, 2020 for International Application No. PCT/US2020/048437.

Watchman FLX™ Left Atrial Appendage Closure Device, Boston Scientific. 5 pages, 2022.

* cited by examiner

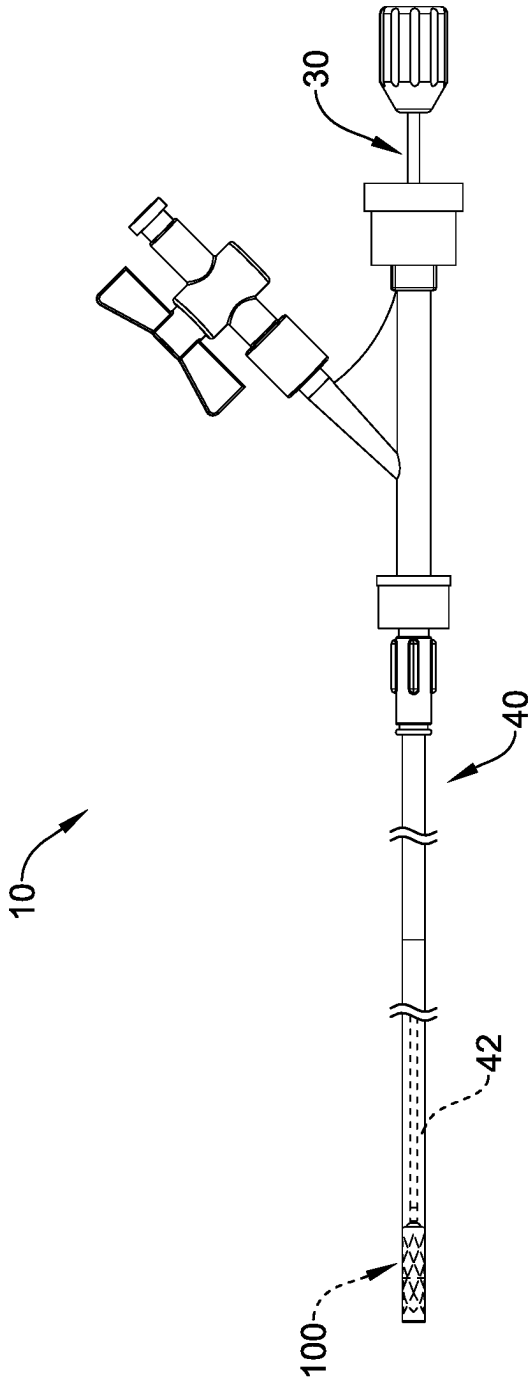


FIG. 1

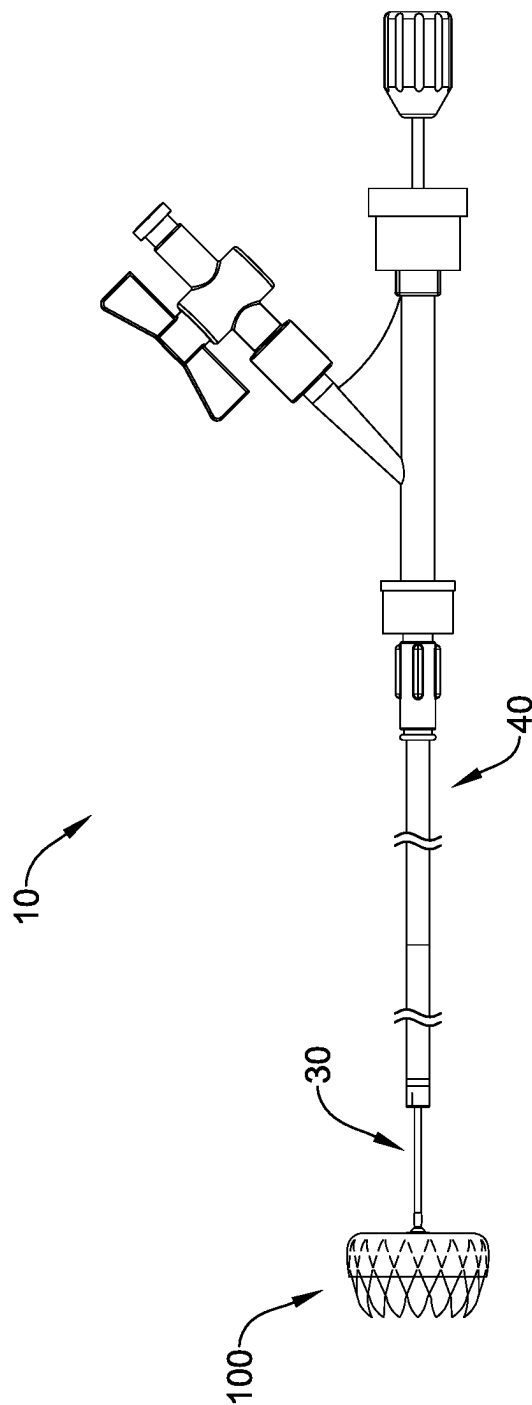


FIG. 2

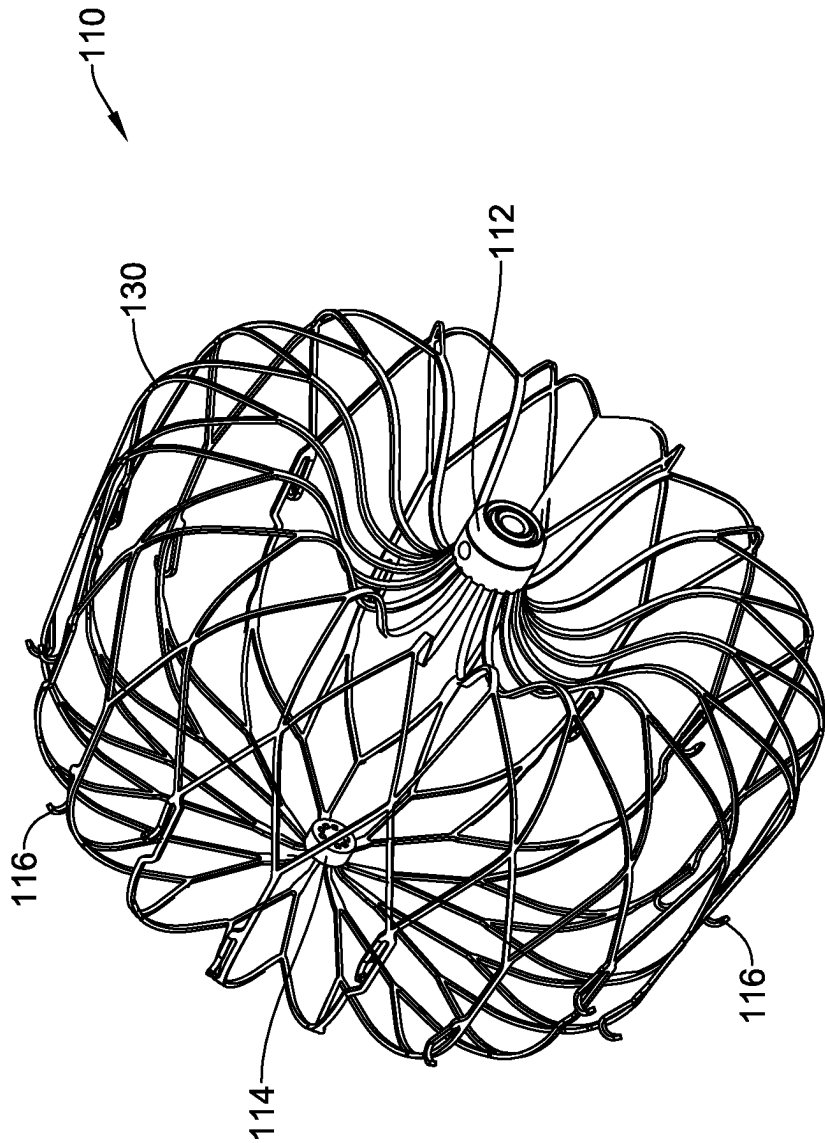


FIG. 3

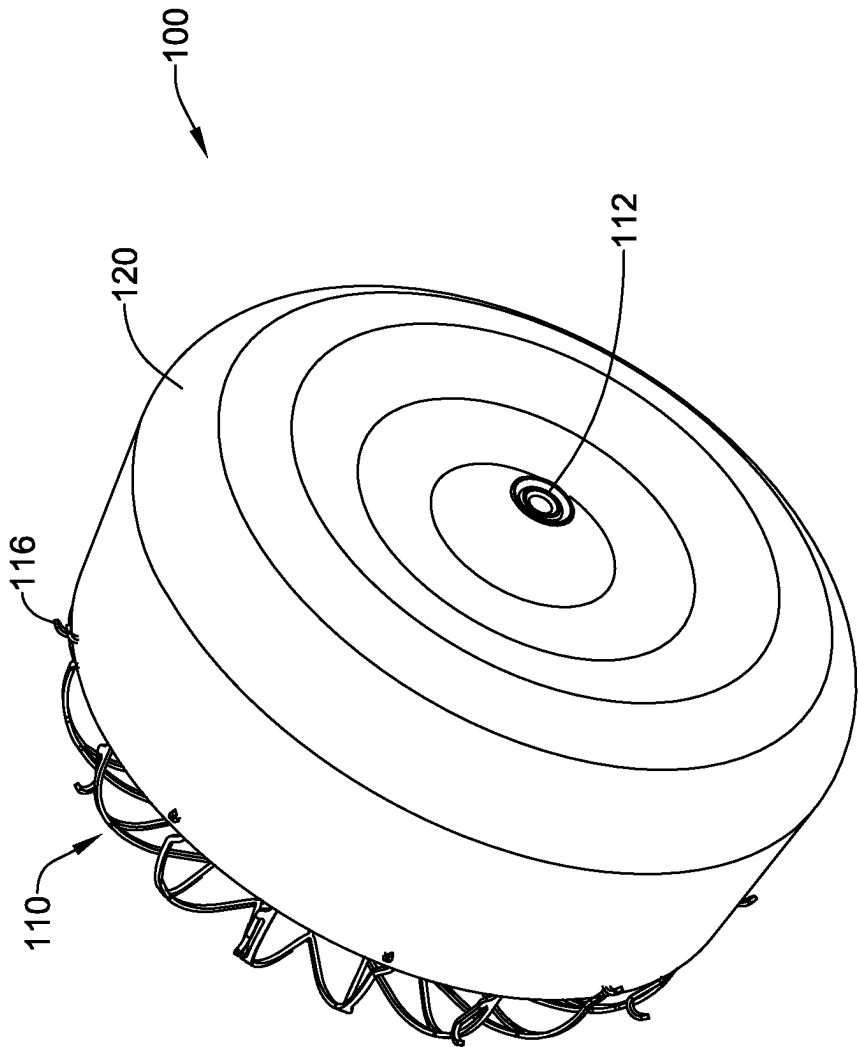


FIG. 4

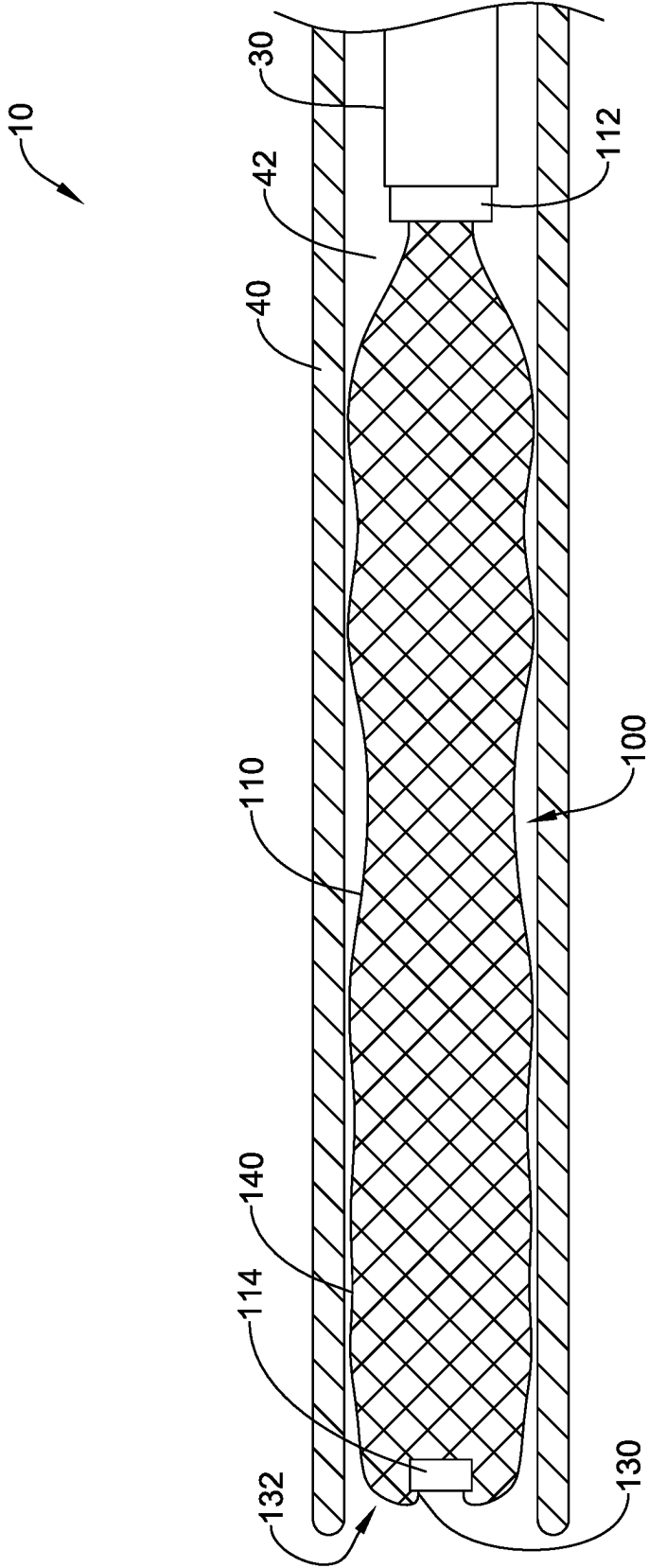


FIG. 5

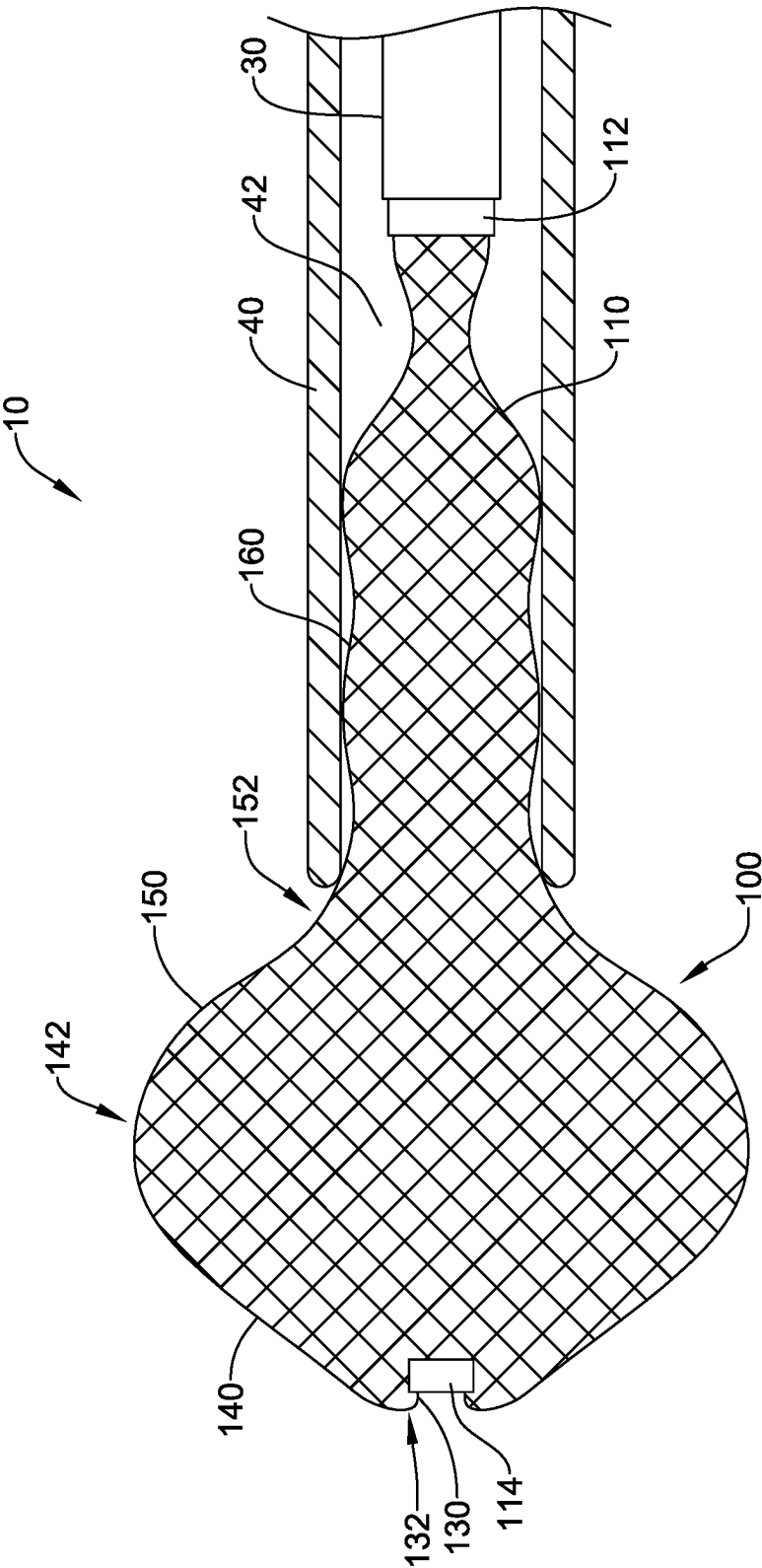


FIG. 6

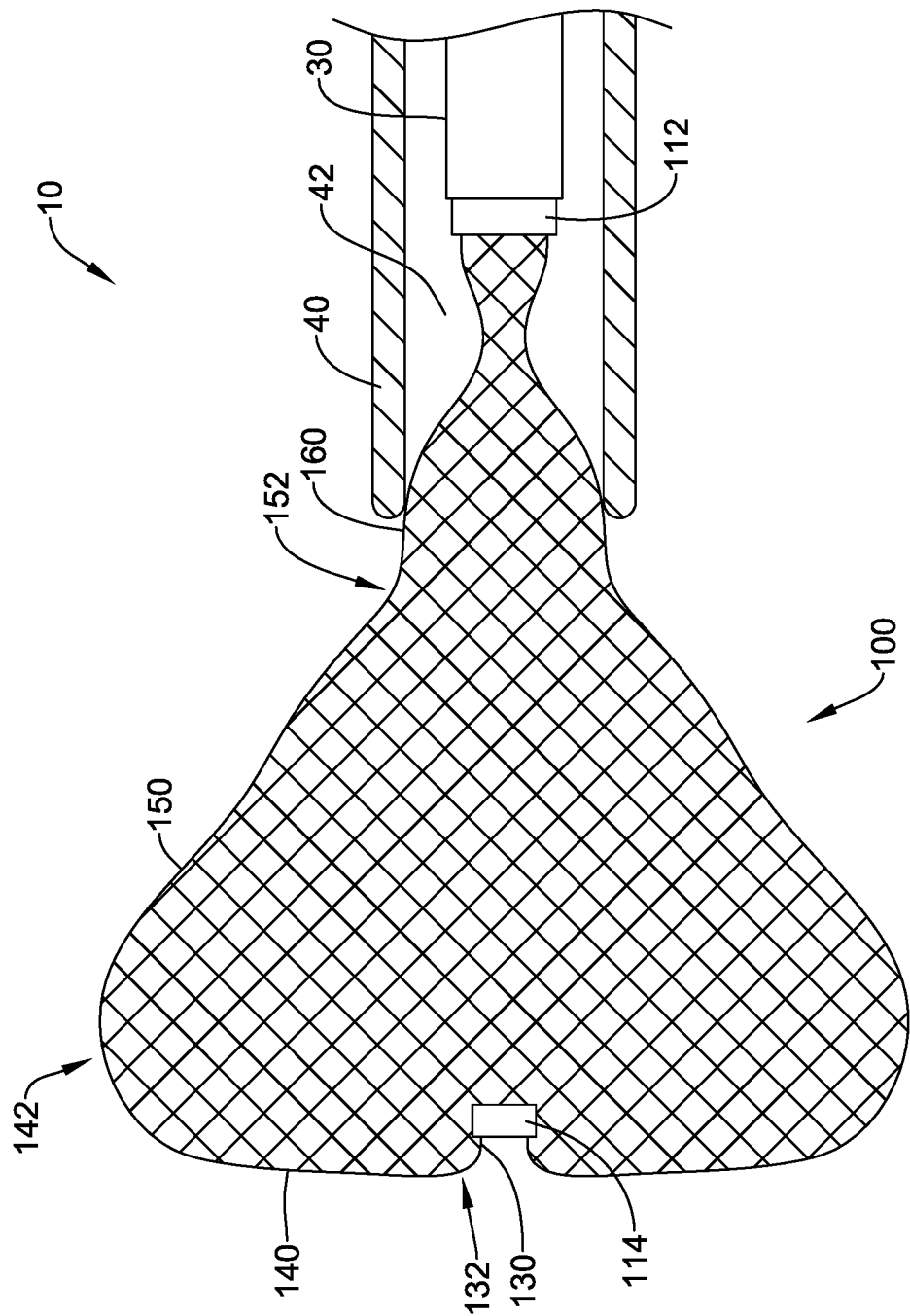


FIG. 7

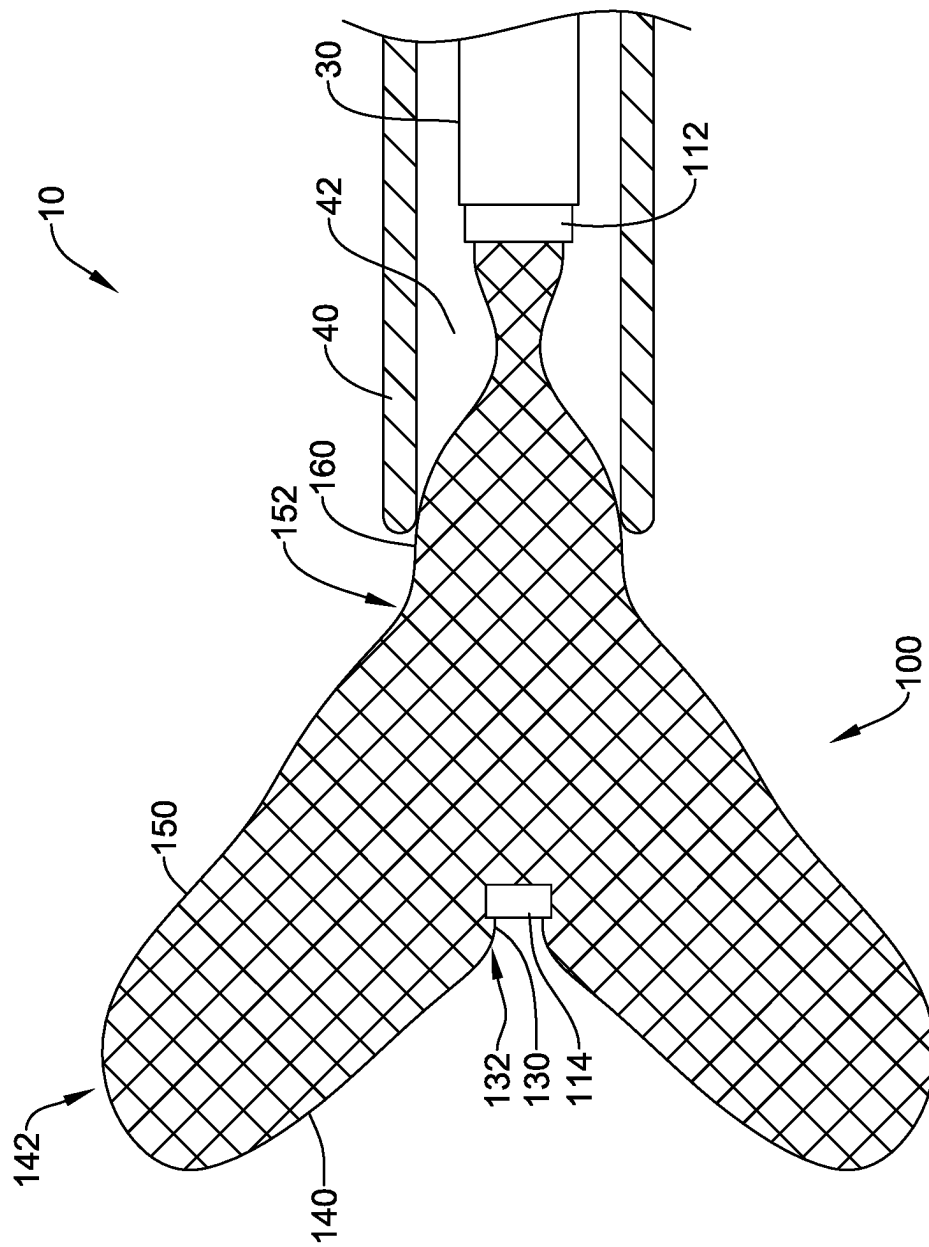
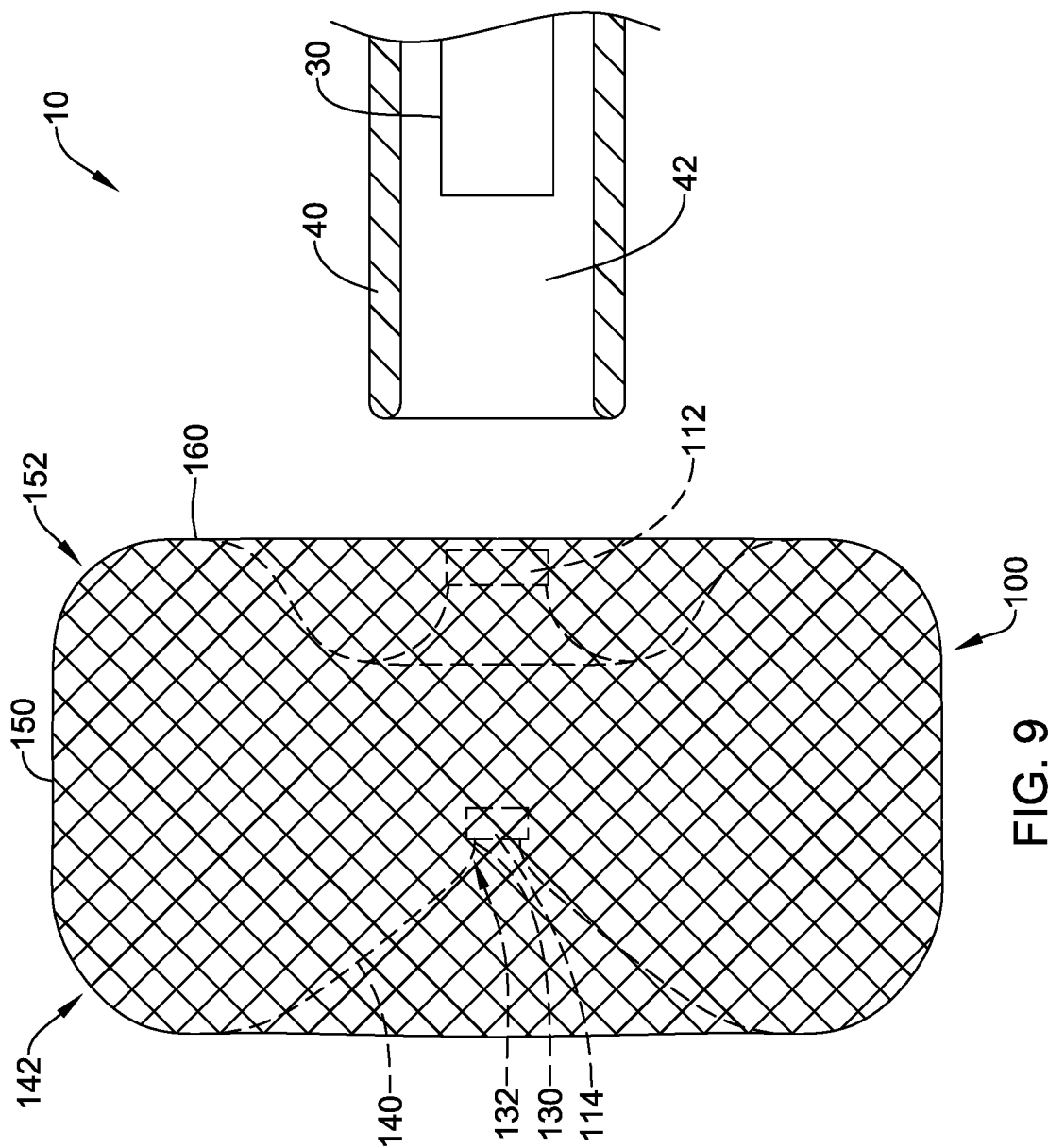


FIG. 8



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LEFT ATRIAL APPENDAGE CLOSURE DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority of U.S. Provisional Application No. 63/219,696 filed Jul. 8, 2021, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

The disclosure relates generally to medical devices and more particularly to medical devices that are adapted for use in percutaneous medical procedures including implantation into the left atrial appendage (LAA) of a heart.

BACKGROUND

The left atrial appendage is a small organ attached to the left atrium of the heart. During normal heart function, as the left atrium constricts and forces blood into the left ventricle, the left atrial appendage constricts and forces blood into the left atrium. The ability of the left atrial appendage to contract assists with improved filling of the left ventricle, thereby playing a role in maintaining cardiac output. However, in patients suffering from atrial fibrillation, the left atrial appendage may not properly contract or empty, causing stagnant blood to pool within its interior, which can lead to the undesirable formation of thrombi within the left atrial appendage.

Thrombi forming in the left atrial appendage may break loose from this area and enter the blood stream. Thrombi that migrate through the blood vessels may eventually plug a smaller vessel downstream and thereby contribute to stroke or heart attack. Clinical studies have shown that the majority of blood clots in patients with atrial fibrillation originate in the left atrial appendage. As a treatment, medical devices have been developed which are deployed to close off the left atrial appendage. Of the known medical devices and methods, each has certain advantages and disadvantages. There is an ongoing need to provide alternative medical devices and introducers as well as alternative methods for manufacturing and using medical devices and introducers.

SUMMARY

In one example, a left atrial appendage closure device may comprise an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub. When the expandable framework is fully constrained in a first position by a delivery sheath, a first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to a first bend and a second segment of the plurality of struts extends from the first bend proximally. A first amount of the expandable framework is unconstrained by the delivery sheath in a second position. In the second position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend proximally and radially outward to a second bend, a third segment of the plurality of struts extends from the second bend proximally and radially inward to a third bend,

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and a fourth segment of the plurality of struts extends from the third bend proximally to within the delivery sheath.

In addition or alternatively to any example disclosed herein, in the second position the first segment and the second segment form an acute angle opening inwardly toward an interior of the expandable framework.

In addition or alternatively to any example disclosed herein, a second amount of the expandable framework greater than the first amount is unconstrained by the delivery sheath in a third position. In the third position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend radially outward generally perpendicular to the central longitudinal axis to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward to the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath.

In addition or alternatively to any example disclosed herein, in the third position the second segment and the third segment form an acute angle opening inwardly toward the central longitudinal axis.

In addition or alternatively to any example disclosed herein, in the third position the third segment and the fourth segment form an obtuse angle opening outwardly away from the central longitudinal axis.

In addition or alternatively to any example disclosed herein, a third amount of the expandable framework greater than the second amount is unconstrained by the delivery sheath in a fourth position. In the fourth position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward toward the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath.

In addition or alternatively to any example disclosed herein, in the fourth position the first segment and the second segment form an obtuse angle opening inwardly toward an interior of the expandable framework.

In addition or alternatively to any example disclosed herein, in the fourth position the second segment and the third segment form an acute angle opening inwardly toward the central longitudinal axis.

In addition or alternatively to any example disclosed herein, in the fourth position the third segment and the fourth segment form an obtuse angle opening outwardly away from the central longitudinal axis.

In addition or alternatively to any example disclosed herein, a fourth amount of the expandable framework greater than the third amount is unconstrained by the delivery sheath in a fifth position. In the fifth position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally to the third bend, and the fourth segment of the plurality of struts extends from the third bend radially inward toward the proximal hub.

In addition or alternatively to any example disclosed herein, in the fifth position the first segment and the second segment form an obtuse angle opening inwardly.

In addition or alternatively to any example disclosed herein, in the fifth position the second segment and the third segment form an acute angle opening inwardly.

In addition or alternatively to any example disclosed herein, in the fifth position the third segment and the fourth segment form an angle of about 90 degrees or less opening inwardly.

In addition or alternatively to any example disclosed herein, the distal hub is disposed proximal of the first bend.

In addition or alternatively to any example disclosed herein, in the second position the distal hub is disposed distal of the second bend.

In addition or alternatively to any example disclosed herein, a left atrial appendage closure device may comprise an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub. As the expandable framework shifts from fully constrained to fully unconstrained, the expandable framework transitions sequentially through a plurality of positions. In a first position, a first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to a first bend and a second segment of the plurality of struts extends from the first bend proximally and generally parallel to the central longitudinal axis. In a second position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally conical shape tapering radially outward in a proximal direction from the first bend toward a second bend.

In addition or alternatively to any example disclosed herein, in the second position, the distal hub is disposed proximal of the first bend and the distal hub is disposed distal of the second bend.

In addition or alternatively to any example disclosed herein, in a third position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally planar shape oriented generally perpendicular to the central longitudinal axis.

In addition or alternatively to any example disclosed herein, in a fourth position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally conical shape tapering radially outward in a distal direction from the first bend toward the second bend.

In addition or alternatively to any example disclosed herein, a left atrial appendage closure device system may comprise a delivery sheath having a lumen extending therein, and a left atrial appendage closure device comprising an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub. When the expandable framework is disposed within the lumen of the delivery sheath in a first position, a first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to a first bend and a second segment of the plurality of struts extends from the first bend proximally. Relative axial translation between the delivery sheath and the expandable framework exposes some of the expandable framework in a second position. In the second position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the

second segment of the plurality of struts extends from the first bend proximally and radially outward to a second bend, a third segment of the plurality of struts extends from the second bend proximally and radially inward to a third bend, and a fourth segment of the plurality of struts extends from the third bend proximally to within the delivery sheath. Relative axial translation between the delivery sheath and the expandable framework exposes more of the expandable framework in a third position than in the second position. In the third position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend radially outward generally perpendicular to the central longitudinal axis to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward to the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath. Relative axial translation between the delivery sheath and the expandable framework exposes more of the expandable framework in a fourth position than in the third position. In the fourth position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward toward the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath. Relative axial translation between the delivery sheath and the expandable framework exposes all of the expandable framework in a fifth position. In the fifth position the first segment of the plurality of struts extends distally from the distal hub parallel to the central longitudinal axis to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally to the third bend, and the fourth segment of the plurality of struts extends from the third bend radially inward toward the proximal hub.

The above summary of some embodiments, aspects, and/or examples is not intended to describe each embodiment or every implementation of the present disclosure. The figures and the detailed description more particularly exemplify aspects of these embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:

FIGS. 1-2 are side views of a left atrial appendage closure device system;

FIG. 3 illustrates selected aspects of a left atrial appendage closure device;

FIG. 4 illustrates selected aspects of the left atrial appendage closure device of FIG. 3; and

FIGS. 5-9 are schematic partial cross-sectional views illustrating selected aspects related to deploying the left atrial appendage closure device of FIGS. 3-4.

While aspects of the disclosure are amenable to various modifications and alternative forms, examples are shown in the drawings and described herein. It should be understood, however, that the intention is not to limit aspects of the

disclosure to the particular embodiments described. On the contrary, the disclosure shall cover all modifications, equivalents, and alternatives falling within the spirit and scope thereof.

DETAILED DESCRIPTION

The following description should be read with reference to the drawings, which are not necessarily to scale, wherein like reference numerals indicate like elements throughout the several views. The detailed description and drawings are intended to illustrate but not limit the present disclosure. Those skilled in the art will recognize that the various elements described and/or shown may be arranged in various combinations and configurations without departing from the scope of the disclosure. The detailed description and drawings illustrate exemplary aspects of the disclosure. However, in the interest of clarity and ease of understanding, while every feature and/or element may not be shown in each drawing, the feature(s) and/or element(s) may be understood to be present regardless, unless otherwise specified.

For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about,” in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (e.g., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.

The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

Although some suitable dimensions, ranges, and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges, and/or values may deviate from those expressly disclosed.

As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise. It is to be noted that in order to facilitate understanding, certain features of the disclosure may be described in the singular, even though those features may be plural or recurring within the disclosed embodiment(s). Each instance of the features may include and/or be encompassed by the singular disclosure(s), unless expressly stated to the contrary. For simplicity and clarity purposes, all elements of the present disclosure are not necessarily shown in each figure or discussed in detail below. However, it will be understood that the following discussion may apply equally to any and/or all of the components for which there are more than one, unless explicitly stated to the contrary. Additionally, not all instances of some elements or features may be shown in each figure for clarity.

Relative terms such as “proximal”, “distal”, “advance”, “retract”, variants thereof, and the like, may be generally considered with respect to the positioning, direction, and/or operation of various elements relative to a user/operator/manipulator of the device, wherein “proximal” and “retract” indicate or refer to closer to or toward the user and “distal” and “advance” indicate or refer to farther from or away from the user. In some instances, the terms “proximal” and “distal” may be arbitrarily assigned in an effort to facilitate understanding of the disclosure, and such instances will be readily apparent to the skilled artisan. Other relative terms, such as “upstream”, “downstream”, “inflow”, and “outflow” refer to a direction of fluid flow within a lumen, such as a body lumen, a blood vessel, or within a device. Still other relative terms, such as “axial”, “circumferential”, “longitudinal”, “lateral”, “radial”, etc. and/or variants thereof generally refer to direction and/or orientation relative to a central longitudinal axis of the disclosed structure or device.

The term “extent” may be understood to mean the greatest measurement of a stated or identified dimension, unless the extent or dimension in question is preceded by or identified as a “minimum”, which may be understood to mean the smallest measurement of the stated or identified dimension. For example, “outer extent” may be understood to mean an outer dimension, “radial extent” may be understood to mean a radial dimension, “longitudinal extent” may be understood to mean a longitudinal dimension, etc. Each instance of an “extent” may be different (e.g., axial, longitudinal, lateral, radial, circumferential, etc.) and will be apparent to the skilled person from the context of the individual usage. Generally, an “extent” may be considered the greatest possible dimension measured according to the intended usage, while a “minimum extent” may be considered the smallest possible dimension measured according to the intended usage. In some instances, an “extent” may generally be measured orthogonally within a plane and/or cross-section, but may be, as will be apparent from the particular context, measured differently—such as, but not limited to, angularly, radially, circumferentially (e.g., along an arc), etc.

The terms “monolithic” and “unitary” shall generally refer to an element or elements made from or consisting of a single structure or base unit/element. A monolithic and/or unitary element shall exclude structure and/or features made by assembling or otherwise joining multiple discrete structures or elements together.

It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect the particular feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

For the purpose of clarity, certain identifying numerical nomenclature (e.g., first, second, third, fourth, etc.) may be used throughout the description and/or claims to name

and/or differentiate between various described and/or claimed features. It is to be understood that the numerical nomenclature is not intended to be limiting and is exemplary only. In some embodiments, alterations of and deviations from previously used numerical nomenclature may be made in the interest of brevity and clarity. That is, a feature identified as a "first" element may later be referred to as a "second" element, a "third" element, etc. or may be omitted entirely, and/or a different feature may be referred to as the "first" element. The meaning and/or designation in each instance will be apparent to the skilled practitioner.

The left atrial appendage may be attached to and in fluid communication with a left atrium of a patient's heart. In some patients, the left atrial appendage may have a complex geometry and/or irregular surface area. Those of skill in the art will also recognize that the medical devices and methods disclosed herein may be adapted for various sizes and shapes of the left atrial appendage, as necessary. The left atrial appendage may include a generally longitudinal axis arranged along a depth of a main body of the left atrial appendage. The main body may include a wall and an ostium forming a proximal mouth. In some embodiments, a lateral extent of the ostium and/or the wall may be smaller or less than a depth of the main body along the longitudinal axis, or a depth of the main body may be greater than a lateral extent of the ostium and/or the wall. In some embodiments, the left atrial appendage may include a tail-like element associated with a distal portion of the main body, which element may protrude radially or laterally away from the main body.

The following figures illustrate selected components and/or arrangements of a left atrial appendage closure device, a left atrial appendage closure device system, and/or methods of using the left atrial appendage closure device and/or the left atrial appendage closure device system. It should be noted that in any given figure, some features may not be shown, or may be shown schematically, for simplicity. Additional details regarding some of the components of the implant and/or the system may be illustrated in other figures in greater detail. While discussed in the context of occluding the left atrial appendage, the left atrial appendage closure device and/or the left atrial appendage closure device system may also be used for other interventions and/or percutaneous medical procedures within a patient. Similarly, the devices and methods described herein with respect to percutaneous deployment may be used in other types of surgical procedures, as appropriate. For example, in some examples, the devices may be used in a non-percutaneous procedure. Devices and methods in accordance with the disclosure may also be adapted and configured for other uses within the anatomy.

FIGS. 1-2 illustrate selected components and/or arrangements of a left atrial appendage closure device system 10 which may be used for occluding a left atrial appendage. It should be noted that in any given figure, some features of the left atrial appendage closure device system 10 may not be shown, or may be shown schematically, for simplicity. Additional details regarding some of the components of the left atrial appendage closure device system 10 may be illustrated in other figures in greater detail.

The left atrial appendage closure device system 10 may include a delivery sheath 40 having a lumen 42 extending from a proximal opening to a distal opening, a core wire 30 slidably disposed within the lumen 42, and a left atrial appendage closure device 100 for occluding the left atrial appendage. The left atrial appendage closure device 100 may include an expandable framework 110 (e.g., FIG. 3) configured to shift between a fully constrained configuration

(e.g., FIG. 1), wherein the left atrial appendage closure device 100 is disposed within the lumen 42 proximate the distal opening in the delivery configuration, and a fully unconstrained configuration (e.g., FIG. 2), wherein the left atrial appendage closure device 100 and/or the expandable framework 110 is configured to shift between the fully constrained configuration and the fully unconstrained configuration as the left atrial appendage closure device 100 is translated relative to the delivery sheath 40. In at least some embodiments, the expandable framework 110 may be self-biased toward the fully unconstrained configuration.

The left atrial appendage closure device 100 may be disposed at and/or releasably securable to a distal portion of the core wire 30. The core wire 30 may be slidably and/or rotatably disposed within the lumen 42 of the delivery sheath 40. In some embodiments, a proximal end of the core wire 30 may extend proximally of a proximal end of the delivery sheath 40 and/or the proximal opening of the lumen 42 for manual manipulation by a clinician or practitioner. In some embodiments, the left atrial appendage closure device 100 may be removably attached, joined, secured, or otherwise connected to a distal end of the core wire 30. The core wire 30 may be configured to and/or may be capable of axially translating the left atrial appendage closure device 100 relative to the delivery sheath 40. In one example, the core wire 30 may be advanced distally while the delivery sheath 40 is held in a constant position. In another example, the core wire 30 may be advanced distally while the delivery sheath 40 is retracted proximally. In yet another example, the core wire 30 may be held in a constant position while the delivery sheath 40 is retracted proximally relative to the core wire 30 and/or the left atrial appendage closure device 100. Other configurations are also contemplated. The delivery sheath 40 and/or the core wire 30 may have a selected level of axial stiffness and/or pushability characteristics while also having a selected level of flexibility to permit navigation through the patient's vasculature.

Some suitable, but non-limiting, examples of materials for the left atrial appendage closure device system 10, the core wire 30, the delivery sheath 40, and/or the left atrial appendage closure device 100, etc. are discussed below. It is contemplated that any exemplary left atrial appendage closure device disclosed herein may be used in accordance with and/or be associated with the example left atrial appendage closure device system 10 described above.

The left atrial appendage closure device 100 may comprise an expandable framework 110 configured to shift axially and/or radially along a central longitudinal axis between the fully constrained configuration and the fully unconstrained configuration. In the fully constrained configuration, the expandable framework 110 may be axially elongated and/or radially compressed. In the fully unconstrained configuration, the expandable framework 110 may be axially shortened and/or radially expanded.

As seen in FIG. 3, which illustrates selected aspects of the left atrial appendage closure device 100 in the fully unconstrained configuration, the expandable framework 110 may have a plurality of struts disposed about the central longitudinal axis. In some embodiments, the plurality of struts may define a plurality of cells. In some embodiments, the plurality of cells may be a plurality of closed cells. In some embodiments, the plurality of cells may be a plurality of open cells. In some embodiments, the plurality of cells may include a plurality of open cells and a plurality of closed cells in various combinations and/or arrangements.

The expandable framework 110 may include a proximal hub 112 and a distal hub 114. In some embodiments, the

proximal hub **112** and/or the distal hub **114** may be centered on and/or coaxial with the longitudinal axis. The plurality of struts may be joined together at and/or fixedly attached to the proximal hub **112** and/or the distal hub **114**. The proximal hub **112** may be configured to releasably connect, secure, and/or attach the left atrial appendage closure device **100** and/or the expandable framework **110** to the core wire **30**. In some embodiments, the proximal hub **112** may include internal threads configured to rotatably and/or threadably engage an externally threaded distal end of the core wire **30**. Other configurations for releasably securing the left atrial appendage closure device **100** to the core wire **30** are also contemplated. As noted herein, some features are not shown in every figure to improve clarity.

The expandable framework **110** and/or the plurality of struts may be formed and/or cut from a tubular member. In some embodiments, the expandable framework **110** and/or the plurality of struts may be integrally formed and/or cut from a unitary member. In some embodiments, the expandable framework **110** and/or the plurality of struts may be integrally formed and/or cut from a unitary tubular member and subsequently formed and/or heat set to a desired shape in the fully unconstrained configuration. In some embodiments, the expandable framework **110** and/or the plurality of struts may be integrally formed and/or cut from a unitary flat member or sheet, and then rolled or formed into a tubular structure and subsequently formed and/or heat set to the desired shape in the fully unconstrained configuration. Some exemplary means and/or methods of making and/or forming the expandable framework **110** and/or the plurality of struts include laser cutting, machining, punching, stamping, electro discharge machining (EDM), chemical dissolution, etc. Other means and/or methods are also contemplated.

As would be understood by the skilled person, anatomical features may vary in size and/or shape. In some embodiments, the left atrial appendage may have an irregular (e.g., elongated and/or oblong) cross-sectional shape. In some embodiments, the expandable framework **110** may be compliant and substantially conform to and/or be in sealing engagement with the shape and/or geometry of a lateral wall of a left atrial appendage when deployed and/or expanded therein. In some embodiments, the left atrial appendage closure device **100** may expand to a size, extent, or shape less than or different from the fully unconstrained configuration, as determined by the surrounding tissue and/or lateral wall of the left atrial appendage. In some embodiments, the expandable framework **110** may be configured to shape and/or stretch the tissue of the left atrial appendage such that the lateral wall of the left atrial appendage substantially conforms to an outer shape of the expandable framework **110**. Other configurations are also contemplated.

In some embodiments, the expandable framework **110** may include at least one anchoring member **116** extending radially outward therefrom in the fully unconstrained configuration. In some embodiments, the expandable framework **110** may include at least one anchoring member **116** extending radially outward from the expandable framework **110**. In some embodiments, the expandable framework **110** may include at least one anchoring member **116** extending radially outward from the expandable framework **110** proximate a proximal shoulder of the expandable framework **110**. In some embodiments, the expandable framework **110** may include at least one anchoring member **116** extending radially outward from the expandable framework **110** proximate a midsection of the expandable framework **110**. In some

body of the left atrial appendage. In some embodiments, the at least one anchoring member **116** may be formed as J-shaped hooks having a free end extending in and/or directed toward a proximal direction with respect to the central longitudinal axis of the left atrial appendage closure device **100** and/or the expandable framework **110**. Other configurations are also contemplated.

In some embodiments, the left atrial appendage closure device **100** may optionally include the occlusive element **120** connected to, disposed on, disposed over, disposed about, and/or disposed radially outward of at least a portion of the expandable framework **110** and/or the plurality of struts, as seen in FIG. 4. In some embodiments, the occlusive element **120** may be attached to the proximal hub **112** and/or may be attached to the expandable framework at the proximal hub **112**. In some embodiments, the occlusive element **120** may extend radially outward from and/or may extend distally from the proximal hub **112**. In some embodiments, the occlusive element **120** may be attached and/or secured to the expandable framework **110** at a plurality of discrete locations. In some embodiments, one of, some of, and/or all of the at least one anchoring member **116** may extend through an occlusive element **120**, where present.

In some embodiments, the occlusive element **120** may include a membrane, a fabric, a mesh, a tissue element, or another suitable construction. In some embodiments, the occlusive element **120** may be porous. In some embodiments, the occlusive element **120** may be non-porous. In some embodiments, the occlusive element **120** may be permeable to selected gases and/or fluids. In some embodiments, the occlusive element **120** may be substantially impermeable to selected gases and/or fluids, such as blood, water, etc. In some embodiments, the occlusive element **120** may be designed, sized, and/or configured to prevent thrombus and/or embolic material from passing out of the left atrial appendage into the left atrium and/or the patient's bloodstream. In some embodiments, the occlusive element **120** may be configured to promote endothelialization after implantation, thereby effectively removing the target site (e.g., the left atrial appendage, etc.) from the patient's circulatory system. Some suitable, but non-limiting, examples of materials for the occlusive element **120** are discussed below.

FIGS. 5-9 schematically illustrate selected aspects of the left atrial appendage closure device **100** and/or the left atrial appendage closure device system **10** during deployment of the left atrial appendage closure device **100**. For clarity and ease of understanding, some elements of the left atrial appendage closure device **100** are not shown but shall be understood to be present in accordance and/or consistent with other figures and/or description of the disclosure. FIGS. 5-8 illustrate the left atrial appendage closure device **100** in partial cross-section. FIG. 9 illustrates the left atrial appendage closure device **100** using broken lines to show hidden features as may be understood from other figures. The occlusive element **120** is not shown in FIG. 9. It may be seen in FIGS. 5-9 that as the expandable framework **110** of the left atrial appendage closure device **100** shifts from fully constrained to fully unconstrained, the expandable framework **110** may transition sequentially through a plurality of positions. In some embodiments, the plurality of positions may include a first position, a second position, a third position, a fourth position, and/or a fifth position as described herein. In some embodiments, the plurality of positions may include additional and/or other positions.

Returning now to FIG. 5, in some embodiments, the expandable framework **110** may be fully constrained in a

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first position by the delivery sheath **40**. In some embodiments, the expandable framework **110** may be disposed within the lumen **42** of the delivery sheath **40** in the first position. In some embodiments, in the first position a first segment **130** of the plurality of struts of the expandable framework **110** may extend distally from the distal hub **114** substantially parallel to the central longitudinal axis to a first bend **132** and a second segment **140** of the plurality of struts of the expandable framework **110** may extend from the first bend **132** proximally. In some embodiments, in the first position the second segment **140** may extend from the first bend **132** proximally and generally parallel to the central longitudinal axis. In the first position, the distal hub **114** may be disposed proximal of the first bend **132**. In some embodiments, the first segment **130** may be secured to the distal hub **114**. In some embodiments, the first segment **130** may be fixedly attached to the distal hub **114**. For example, the first segment **130** may be welded, adhesively bonded, etc. to the distal hub **114**. Other configurations are also contemplated.

In some embodiments, relative axial translation between the delivery sheath **40** and the expandable framework **110** may expose some of the expandable framework **110** and/or the plurality of struts in a second position, as seen in FIG. 6. In some embodiments, a first amount of the expandable framework **110** and/or the plurality of struts may be exposed from and/or unconstrained by the delivery sheath **40** in the second position. In some embodiments, the first amount may be less than about 55% of an axial length, a volume, a weight, and/or a surface area of the expandable framework **110** and/or the plurality of struts. In some embodiments, the first amount may be about 15% to about 55% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. Other configurations and/or ranges are also contemplated. In some embodiments, in the second position, the expandable framework **110** may have a maximum radial extent of about 6 millimeters (mm) to about 10 mm. In some embodiments, in the second position, the expandable framework **110** may have a maximum radial extent of about 8 mm. In some embodiments, in the second position, the maximum radial extent of the expandable framework **110** may be about twice or about 200% of a maximum outer extent of a distal end of the delivery sheath **40**. Other configurations and/or sizes are also contemplated.

In some embodiments, in the second position the first segment **130** of the plurality of struts of the expandable framework **110** may extend distally from the distal hub **114** substantially parallel to the central longitudinal axis to the first bend **132**. In the second position, the second segment **140** of the plurality of struts of the expandable framework **110** may extend from the first bend **132** proximally and radially outward toward and/or to a second bend **142**. In the second position, a third segment **150** of the plurality of struts of the expandable framework **110** may extend from the second bend **142** proximally and radially inward toward and/or to a third bend **152**. In some embodiments, in the second position, a fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** proximally to within the lumen **42** of the delivery sheath **40**. In some embodiments, in the second position, the fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** proximally toward the proximal hub **112** disposed within the lumen **42** of the delivery sheath **40**.

In some embodiments, in the second position, the distal hub **114** is disposed proximal of the first bend **132**. In some embodiments, in the second position, the distal hub **114** may

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be disposed distal of the second bend **142**. For example, in the second position, the distal hub **114** may be positioned distal of a plane extending through, including at least a portion of, and/or tangent to the second bend **142** and oriented perpendicular to the central longitudinal axis. In some embodiments, in the second position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may form and/or intersect to form an acute angle opening inwardly toward an interior of the expandable framework **110**. In some embodiments, in the second position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may form and/or intersect to form an acute angle opening radially outwardly from the central longitudinal axis. In some embodiments, in the second position, the second segment **140** of the plurality of struts of the expandable framework **110**, if swept circumferentially around the central longitudinal axis, may define a generally conical shape tapering radially outward in a proximal direction from the first bend **132** toward the second bend **142**.

In some embodiments, relative axial translation between the delivery sheath **40** and the expandable framework **110** may expose more of the expandable framework **110** and/or the plurality of struts in a third position than in the second position, as seen in FIG. 7. In some embodiments, a second amount of the expandable framework **110** and/or the plurality of struts greater than the first amount may be exposed from and/or unconstrained by the delivery sheath **40** in the third position. In some embodiments, the second amount may be less than about 75% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. In some embodiments, the second amount may be about 40% to about 75% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. Other configurations and/or ranges are also contemplated.

In some embodiments, in the third position the first segment **130** of the plurality of struts of the expandable framework **110** may extend distally from the distal hub **114** substantially parallel to the central longitudinal axis to the first bend **132**. In the third position, the second segment **140** of the plurality of struts of the expandable framework **110** may extend from the first bend **132** radially outward generally perpendicular to the central longitudinal axis toward and/or to the second bend **142**. In the third position, the third segment **150** of the plurality of struts of the expandable framework **110** may extend from the second bend **142** proximally and radially inward toward and/or to the third bend **152**. In some embodiments, in the third position, the fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** proximally to within the lumen **42** of the delivery sheath **40**. In some embodiments, in the third position, the fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** proximally toward the proximal hub **112** disposed within the lumen **42** of the delivery sheath **40**.

In some embodiments, in the third position, the distal hub **114** is disposed proximal of the first bend **132**. In some embodiments, in the third position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may form and/or intersect to form a generally right angle opening toward the interior of the expandable framework **110**. In some embodiments, in the third position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may

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form and/or intersect to form a generally right angle opening radially outwardly away from the central longitudinal axis. In some embodiments, in the third position, the second segment **140** of the plurality of struts and the third segment **150** of the plurality of struts may form and/or intersect to form an acute angle opening radially inwardly toward the interior of the expandable framework **110**. In some embodiments, in the third position, the second segment **140** of the plurality of struts and the third segment **150** of the plurality of struts may form and/or intersect to form an acute angle opening radially inwardly toward the central longitudinal axis. In some embodiments, in the third position, the third segment **150** of the plurality of struts and the fourth segment **160** of the plurality of struts may form and/or intersect to form an obtuse angle opening radially outwardly away from the central longitudinal axis. In some embodiments, in the third position, the second segment **140** of the plurality of struts of the expandable framework **110**, if swept circumferentially around the central longitudinal axis, may define a generally planar shape oriented generally perpendicular to the central longitudinal axis.

In some embodiments, relative axial translation between the delivery sheath **40** and the expandable framework **110** may expose more of the expandable framework **110** and/or the plurality of struts in a fourth position than in the third position, as seen in FIG. **8**. In some embodiments, a third amount of the expandable framework **110** and/or the plurality of struts greater than the second amount may be exposed from and/or unconstrained by the delivery sheath **40** in the fourth position. In some embodiments, the third amount may be less than about 95% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. In some embodiments, the third amount may be about 60% to about 95% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. Other configurations and/or ranges are also contemplated.

In some embodiments, in the fourth position the first segment **130** of the plurality of struts of the expandable framework **110** may extend distally from the distal hub **114** substantially parallel to the central longitudinal axis to the first bend **132**. In the fourth position, the second segment **140** of the plurality of struts of the expandable framework **110** may extend from the first bend **132** distally and radially outward toward and/or to the second bend **142**. In the fourth position, the third segment **150** of the plurality of struts of the expandable framework **110** may extend from the second bend **142** proximally and radially inward toward and/or to the third bend **152**. In some embodiments, in the fourth position, the fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** proximally toward the proximal hub **112** disposed within the lumen **42** of the delivery sheath **40**.

In some embodiments, in the fourth position, the distal hub **114** is disposed proximal of the first bend **132**. In some embodiments, in the fourth position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may form and/or intersect to form an obtuse angle opening inwardly toward the interior of the expandable framework **110**. In some embodiments, in the fourth position, the second segment **140** of the plurality of struts and the third segment **150** of the plurality of struts may form and/or intersect to form an acute angle opening radially inwardly toward the central longitudinal axis. In some embodiments, in the fourth position, the third segment **150** of the plurality of struts and the fourth segment **160** of the

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plurality of struts may form and/or intersect to form an obtuse angle opening radially outwardly away from the central longitudinal axis. In some embodiments, in the fourth position, the second segment **140** of the plurality of struts of the expandable framework **110**, if swept circumferentially around the central longitudinal axis, may define a generally conical shape tapering radially outward in a distal direction from the first bend **132** toward the second bend **142**.

In some embodiments, relative axial translation between the delivery sheath **40** and the expandable framework **110** may expose substantially all of the expandable framework **110** and/or the plurality of struts in a fifth position, as seen in FIG. **9**. In some embodiments, a fourth amount of the expandable framework **110** and/or the plurality of struts greater than the third amount may be exposed from and/or unconstrained by the delivery sheath **40** in the fifth position. In some embodiments, the fourth amount may be more than about 95% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. In some embodiments, the fourth amount may be about 100% of the axial length, the volume, the weight, and/or the surface area of the expandable framework **110** and/or the plurality of struts. Other configurations and/or ranges are also contemplated.

In some embodiments, the expandable framework **110** may be in the fully unconstrained configuration in the fifth position. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 16 millimeters (mm) to about 40 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 16 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 20 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 25 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 30 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 35 mm. In some embodiments, in the fully unconstrained configuration, the expandable framework may have a maximum radial extent of about 40 mm. Other configurations and/or sizes are also contemplated.

In some embodiments, in the fifth position the first segment **130** of the plurality of struts of the expandable framework **110** may extend distally from the distal hub **114** substantially parallel to the central longitudinal axis to the first bend **132**. In the fifth position, the second segment **140** of the plurality of struts of the expandable framework **110** may extend from the first bend **132** distally and radially outward toward and/or to the second bend **142**. In the fifth position, the third segment **150** of the plurality of struts of the expandable framework **110** may extend from the second bend **142** proximally and toward and/or to the third bend **152**. In some embodiments, in the fifth position, the third segment **150** of the plurality of struts of the expandable framework **110** may extend from the second bend **142** proximally generally parallel to the central longitudinal axis toward and/or to the third bend **152**. In some embodiments, in the fifth position, the fourth segment **160** of the plurality of struts of the expandable framework **110** may extend from the third bend **152** radially inward toward and/or to the proximal hub **112**.

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In some embodiments, in the fifth position, the distal hub **114** is disposed proximal of the first bend **132**. In some embodiments, in the fifth position, the first segment **130** of the plurality of struts and the second segment **140** of the plurality of struts may form and/or intersect to form an obtuse angle opening inwardly toward the interior of the expandable framework **110**. In some embodiments, in the fifth position, the second segment **140** of the plurality of struts and the third segment **150** of the plurality of struts may form and/or intersect to form an acute angle opening radially inwardly toward the central longitudinal axis. In some embodiments, in the fifth position, the third segment **150** of the plurality of struts and the fourth segment **160** of the plurality of struts may form and/or intersect to form an angle of about 90 degrees or less opening radially inwardly toward the interior of the expandable framework **110**. In some embodiments, in the fifth position, the third segment **150** of the plurality of struts and the fourth segment **160** of the plurality of struts may form and/or intersect to form an angle of about 90 degrees or less opening radially inwardly toward the central longitudinal axis.

A method for occluding the left atrial appendage may comprise advancing the left atrial appendage closure device **100** into the left atrial appendage of the patient's heart. For example, the left atrial appendage closure device **100** may be advanced to the left atrial appendage within the lumen **42** of the delivery sheath **40** in the fully constrained configuration. The method may include deploying the expandable framework **110** from the delivery sheath **40** within the left atrial appendage. The method may further include expanding and/or shifting the expandable framework **110** from the fully constrained configuration toward the fully unconstrained configuration within the left atrial appendage.

As the expandable framework **110** shifts from fully constrained to fully unconstrained, the expandable framework **110** may transition sequentially through a plurality of positions, as described herein. In some embodiments, in the second position, the expandable framework **110** may be moved and/or navigated within the patient's heart, the left atrium, and/or the left atrial appendage. In the second position, the expandable framework **110** may form a generally rounded atraumatic shape. In some embodiments, in the second position, the maximum radial extent of the expandable framework **110** may be about twice or about 200% of a maximum outer extent of a distal end of the delivery sheath **40**. In some embodiments, in the second position, the maximum radial extent of the expandable framework **110** may be about 8 millimeters. Other configurations and/or sizes are also contemplated. In some embodiments, a physician may use the generally rounded atraumatic shape as a navigational tool within the patient's anatomy.

In and/or near the fully unconstrained configuration (e.g., the fifth position), the expandable framework **110** may be urged into contact with, engaged with, and/or anchored to the lateral wall of the main body of the left atrial appendage. In some embodiments, the expandable framework **110** may not fully achieve the fully unconstrained configuration (e.g., the fifth position) due to contact with the wall(s) of the left atrial appendage. However, the inverted shape of the distal portion of the expandable framework **110** (e.g., the first segment **130**, the first bend **132**, the second segment **140**, and the second bend **142**) may prevent the expandable framework **110** from elongating distally, which may compromise locating the left atrial appendage closure device **100** and/or the expandable framework **110** within the left atrial appendage, sealing of the left atrial appendage closure

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device **100** and/or the expandable framework **110** with respect to the left atrial appendage, and/or anchoring of the left atrial appendage closure device **100** and/or the expandable framework **110** within the left atrial appendage. Instead, compressive force applied against the third segment **150** and/or the second bend **142** may urge the distal hub **114** proximally toward the proximal hub **112**. As such, the final shape of the left atrial appendage closure device **100** and/or the expandable framework **110** may be more predictable and locating and anchoring of the left atrial appendage closure device **100** and/or the expandable framework **110** within the left atrial appendage may be improved.

In at least some embodiments, the left atrial appendage closure device **100** and/or the expandable framework **110** may span across the ostium of the left atrial appendage. In some embodiments, the left atrial appendage closure device **100** and/or the expandable framework **110** may span completely across the ostium of the left atrial appendage, thereby effectively removing the left atrial appendage from the circulatory system of the patient.

When satisfied with the positioning of the left atrial appendage closure device **100** within the left atrial appendage, the core wire **30** may be disconnected from the left atrial appendage closure device **100**, thereby leaving the left atrial appendage closure device **100** disposed at and/or in the left atrial appendage. In some embodiments, disconnecting the core wire **30** from the left atrial appendage closure device **100** may include rotating the externally threaded distal end of the core wire **30** relative to the left atrial appendage closure device **100** and/or the proximal hub **112** to disengage the core wire **30** from the left atrial appendage closure device **100**.

In some embodiments, the delivery sheath **40** and/or the core wire **30** may include a keying structure configured to prevent rotation of the core wire **30** relative to the proximal hub **112**. In such embodiments, the keying structure is disengaged prior to rotating the core wire **30** relative to the left atrial appendage closure device **100** and/or the proximal hub **112**. When the keying structure is engaged, rotation of the core wire **30** may be transmitted to left atrial appendage closure device **100** and/or the expandable framework **110**. In some embodiments, rotation of the left atrial appendage closure device **100** and/or the expandable framework **110** may facilitate positioning and/or orientation of the left atrial appendage closure device **100** and/or the expandable framework **110** relative to the left atrial appendage, for example, with respect to an asymmetrical and/or irregular ostium and/or left atrial appendage. Other configurations, purposes, and/or results are also contemplated.

The materials that can be used for the various components of the system (and/or other elements disclosed herein) and the various components thereof disclosed herein may include those commonly associated with medical devices and/or systems. For simplicity purposes, the following discussion refers to the system. However, this is not intended to limit the devices and methods described herein, as the discussion may be applied to other elements, members, components, or devices disclosed herein, such as, but not limited to, the left atrial appendage closure device, the delivery sheath, the core wire, the expandable framework, the occlusive element, etc. and/or elements or components thereof.

In some embodiments, the system and/or components thereof may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material.

Some examples of suitable metals and metal alloys include stainless steel, such as 444V, 444L, and 314LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; combinations thereof; and the like; or any other suitable material.

As alluded to herein, within the family of commercially available nickel-titanium or nitinol alloys, is a category designated “linear elastic” or “non-super-elastic” which, although may be similar in chemistry to conventional shape memory and super-elastic varieties, may exhibit distinct and useful mechanical properties. Linear elastic and/or non-super-elastic nitinol may be distinguished from super elastic nitinol in that the linear elastic and/or non-super-elastic nitinol does not display a substantial “superelastic plateau” or “flag region” in its stress/strain curve like super elastic nitinol does. Instead, in the linear elastic and/or non-super-elastic nitinol, as recoverable strain increases, the stress continues to increase in a substantially linear, or a somewhat, but not necessarily entirely linear relationship until plastic deformation begins or at least in a relationship that is more linear than the super elastic plateau and/or flag region that may be seen with super elastic nitinol. Thus, for the purposes of this disclosure linear elastic and/or non-super-elastic nitinol may also be termed “substantially” linear elastic and/or non-super-elastic nitinol. In some cases, linear elastic and/or non-super-elastic nitinol may also be distinguishable from super-elastic nitinol in that linear elastic and/or non-super-elastic nitinol may accept up to about 2-5% strain while remaining substantially elastic (e.g., before plastically deforming) whereas super-elastic nitinol may accept up to about 8% strain before plastically deforming. Both of these materials can be distinguished from other linear elastic materials such as stainless steel (that can also be distinguished based on its composition), which may accept only about 0.2 to 0.44 percent strain before plastically deforming.

In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy is an alloy that does not show any martensite/austenite phase changes that are detectable by differential scanning calorimetry (DSC) and dynamic metal thermal analysis (DMTA) analysis over a large temperature range. For example, in some embodiments, there may be no martensite/austenite phase changes detectable by DSC and DMTA analysis in the range of about -60 degrees Celsius (° C.) to about 120° C. in the linear elastic and/or non-super-elastic nickel-titanium alloy. The mechanical bending properties of such material may therefore be generally inert to the effect of temperature over this very broad range of temperature. In some embodiments, the mechanical bending properties of the linear elastic and/or non-super-elastic nickel-titanium alloy at ambient or room temperature

are substantially the same as the mechanical properties at body temperature, for example, in that they do not display a super-elastic plateau and/or flag region. In other words, across a broad temperature range, the linear elastic and/or non-super-elastic nickel-titanium alloy maintains its linear elastic and/or non-super-elastic characteristics and/or properties.

In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy may be in the range of about 50 to about 60 weight percent nickel, with the remainder being essentially titanium. In some embodiments, the composition is in the range of about 54 to about 57 weight percent nickel. One example of a suitable nickel-titanium alloy is FHP-NT alloy commercially available from Furukawa Techno Material Co. of Kanagawa, Japan. Other suitable materials may include ULTANIUM™ (available from Neo-Metrics) and GUM METAL™ (available from Toyota). In some other embodiments, a superelastic alloy, for example a superelastic nitinol can be used to achieve desired properties.

In at least some embodiments, portions or all of the system and/or other elements disclosed herein may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids a user in determining the location of the system and/or other elements disclosed herein. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the system and/or other elements disclosed herein to achieve the same result.

In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the system and/or other elements disclosed herein. For example, the system and/or components or portions thereof may be made of a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The system or portions thereof, may also be made from a material that the MM machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nitinol, and the like, and others.

In some embodiments, the system and/or other elements disclosed herein may be made from or include a polymer or other suitable material. Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name

PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), MARLEX® high-density polyethylene, MARLEX® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro(propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-*b*-isobutylene-*b*-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP). For example, the mixture can contain up to about 6 percent LCP.

In some embodiments, the system and/or other elements disclosed herein may include a fabric material disposed over or within the structure. The fabric material may be composed of a biocompatible material, such a polymeric material or biomaterial, adapted to promote tissue ingrowth. In some embodiments, the fabric material may include a bioabsorbable material. Some examples of suitable fabric materials include, but are not limited to, polyethylene glycol (PEG), nylon, polytetrafluoroethylene (PTFE, ePTFE), a polyolefinic material such as a polyethylene, a polypropylene, polyester, polyurethane, and/or blends or combinations thereof.

In some embodiments, the system and/or other elements disclosed herein may include and/or be formed from a textile material. Some examples of suitable textile materials may include synthetic yarns that may be flat, shaped, twisted, textured, pre-shrunk or un-shrunk. Synthetic biocompatible yarns suitable for use in the present disclosure include, but are not limited to, polyesters, including polyethylene terephthalate (PET) polyesters, polypropylenes, polyethylenes, polyurethanes, polyolefins, polyvinyls, polymethylacetates, polyamides, naphthalene dicarboxylene derivatives, natural silk, and polytetrafluoroethylenes. Moreover, at least one of the synthetic yarns may be a metallic yarn or a glass or ceramic yarn or fiber. Useful metallic yarns include those yarns made from or containing stainless steel, platinum, gold, titanium, tantalum or a Ni—Co—Cr-based alloy. The yarns may further include carbon, glass or ceramic fibers. Desirably, the yarns are made from thermoplastic materials including, but not limited to, polyesters, polypropylenes, polyethylenes, polyurethanes, polynaphthalenes, polytetrafluoroethylenes, and the like. The yarns may be of the multifilament, monofilament, or spun types. The type and denier of the yarn chosen may be selected in a manner which forms a biocompatible and implantable prosthesis and, more particularly, a vascular structure having desirable properties.

In some embodiments, the system and/or other elements disclosed herein may include and/or be treated with a suitable therapeutic agent. Some examples of suitable therapeutic agents may include anti-thrombogenic agents (such as heparin, heparin derivatives, urokinase, and PPack (dextro-phenylalanine proline arginine chloromethylketone)); anti-proliferative agents (such as enoxaparin, angiostatin, monoclonal antibodies capable of blocking smooth muscle cell proliferation, hirudin, and acetylsalicylic acid); anti-inflammatory agents (such as dexamethasone, prednisolone, cor-

ticosterone, budesonide, estrogen, sulfasalazine, and mesalamine); antineoplastic/antiproliferative/anti-mitotic agents (such as paclitaxel, 5-fluorouracil, cisplatin, vinblastine, vincristine, epothilones, endostatin, angiostatin and thymidine kinase inhibitors); anesthetic agents (such as lidocaine, bupivacaine, and ropivacaine); anti-coagulants (such as D-Phe-Pro-Arg chloromethyl ketone, an RGD peptide-containing compound, heparin, anti-thrombin compounds, platelet receptor antagonists, anti-thrombin antibodies, anti-platelet receptor antibodies, aspirin, prostaglandin inhibitors, platelet inhibitors, and tick antiplatelet peptides); vascular cell growth promoters (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional activators, and translational promoters); vascular cell growth inhibitors (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional repressors, translational repressors, replication inhibitors, inhibitory antibodies, antibodies directed against growth factors, bifunctional molecules consisting of a growth factor and a cytotoxin, bifunctional molecules consisting of an antibody and a cytotoxin); cholesterol-lowering agents; vasodilating agents; and agents which interfere with endogenous vasoactive mechanisms.

It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps, without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The disclosure's scope is, of course, defined in the language in which the appended claims are expressed.

What is claimed is:

1. A left atrial appendage closure device, comprising:
 - an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub;
 - wherein when the expandable framework is fully constrained in a first position by a delivery sheath, a first segment of the plurality of struts extends distally from the distal hub to a first bend and a second segment of the plurality of struts extends from the first bend proximally;
 - wherein a first amount of the expandable framework is unconstrained by the delivery sheath in a second position;
 - wherein in the second position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend proximally and radially outward to a second bend, a third segment of the plurality of struts extends from the second bend proximally and radially inward to a third bend, and a fourth segment of the plurality of struts extends from the third bend proximally to within the delivery sheath;
 - wherein a second amount of the expandable framework greater than the first amount is unconstrained by the delivery sheath in a third position;
 - wherein in the third position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend radially outward generally perpendicular to the central longitudinal axis to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward to the third bend, and the fourth seg-

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ment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath;

wherein a third amount of the expandable framework greater than the second amount is unconstrained by the delivery sheath in a fourth position;

wherein in the fourth position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward toward the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath.

2. The left atrial appendage closure device of claim 1, wherein in the second position the first segment and the second segment form an acute angle opening inwardly toward an interior of the expandable framework.

3. The left atrial appendage closure device of claim 1, wherein in the third position the second segment and the third segment form an acute angle opening inwardly toward the central longitudinal axis.

4. The left atrial appendage closure device of claim 1, wherein in the third position the third segment and the fourth segment form an obtuse angle opening outwardly away from the central longitudinal axis.

5. The left atrial appendage closure device of claim 1, wherein in the fourth position the first segment and the second segment form an obtuse angle opening inwardly toward an interior of the expandable framework.

6. The left atrial appendage closure device of claim 1, wherein in the fourth position the second segment and the third segment form an acute angle opening inwardly toward the central longitudinal axis.

7. The left atrial appendage closure device of claim 1, wherein in the fourth position the third segment and the fourth segment form an obtuse angle opening outwardly away from the central longitudinal axis.

8. The left atrial appendage closure device of claim 1, wherein a fourth amount of the expandable framework greater than the third amount is unconstrained by the delivery sheath in a fifth position;

wherein in the fifth position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally to the third bend, and the fourth segment of the plurality of struts extends from the third bend radially inward toward the proximal hub.

9. The left atrial appendage closure device of claim 8, wherein in the fifth position the first segment and the second segment form an obtuse angle opening inwardly.

10. The left atrial appendage closure device of claim 8, wherein in the fifth position the second segment and the third segment form an acute angle opening inwardly.

11. The left atrial appendage closure device of claim 8, wherein in the fifth position the third segment and the fourth segment form an angle of about 90 degrees or less opening inwardly.

12. The left atrial appendage closure device of claim 1, wherein the distal hub is disposed proximal of the first bend.

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13. The left atrial appendage closure device of claim 1, wherein in the second position the distal hub is disposed distal of the second bend.

14. A left atrial appendage closure device, comprising:
an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub;

wherein as the expandable framework shifts from fully constrained to fully unconstrained, the expandable framework transitions sequentially through a plurality of positions;

wherein in a first position, a first segment of the plurality of struts extends distally from the distal hub to a first bend and a second segment of the plurality of struts extends from the first bend proximally;

wherein in a second position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally conical shape tapering radially outward in a proximal direction from the first bend toward a second bend;

wherein in a third position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally planar shape oriented generally perpendicular to the central longitudinal axis; and

wherein in a fourth position, the second segment of the plurality of struts, if swept circumferentially around the central longitudinal axis, defines a generally conical shape tapering radially outward in a distal direction from the first bend toward the second bend.

15. The left atrial appendage closure device of claim 14, wherein in the second position, the distal hub is disposed proximal of the first bend and the distal hub is disposed distal of the second bend.

16. A left atrial appendage closure device system, comprising:

a delivery sheath having a lumen extending therein; and
a left atrial appendage closure device comprising an expandable framework having a plurality of struts disposed about a central longitudinal axis, the plurality of struts being joined together at a proximal hub and a distal hub;

wherein when the expandable framework is disposed within the lumen of the delivery sheath in a first position, a first segment of the plurality of struts extends distally from the distal hub to a first bend and a second segment of the plurality of struts extends from the first bend proximally;

wherein relative axial translation between the delivery sheath and the expandable framework exposes some of the expandable framework in a second position;

wherein in the second position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend proximally and radially outward to a second bend, a third segment of the plurality of struts extends from the second bend proximally and radially inward to a third bend, and a fourth segment of the plurality of struts extends from the third bend proximally to within the delivery sheath;

wherein relative axial translation between the delivery sheath and the expandable framework exposes more of the expandable framework in a third position than in the second position;

wherein in the third position the first segment of the plurality of struts extends distally from the distal hub to

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the first bend, the second segment of the plurality of struts extends from the first bend radially outward generally perpendicular to the central longitudinal axis to the second bend, the third segment of the plurality of struts extends from the second bend proximally and radially inward to the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath;

wherein relative axial translation between the delivery sheath and the expandable framework exposes more of the expandable framework in a fourth position than in the third position;

wherein in the fourth position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the

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plurality of struts extends from the second bend proximally and radially inward toward the third bend, and the fourth segment of the plurality of struts extends from the third bend proximally toward the proximal hub disposed within the delivery sheath;

wherein relative axial translation between the delivery sheath and the expandable framework exposes all of the expandable framework in a fifth position;

wherein in the fifth position the first segment of the plurality of struts extends distally from the distal hub to the first bend, the second segment of the plurality of struts extends from the first bend distally and radially outward to the second bend, the third segment of the plurality of struts extends from the second bend proximally to the third bend, and the fourth segment of the plurality of struts extends from the third bend radially inward toward the proximal hub.

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